

Classical Homotopy Theory

Labix

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Abstract

- Notes on Algebraic Topology by Oscar Randal-Williams

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1 A Convenient Category of Spaces

Reason:

- Want $- \wedge -$ associative and unital and commutative (so that the category is symmetric monoidal)
- Want adjunction $- \times X$ and $\text{Hom}_{\mathcal{C}}(X, -)$ (non pointed)
- Want adjunction $- \wedge X$ and $\text{Map}_*(X, -)$ (pointed) (Intuitively, $X \wedge Y$ represents maps from $X \times Y$ that are base point preserving separately in each variable)

1.1 Compactly Generated Spaces

Definition 1.1.1 (Compactly Generated Spaces) Let X be a space. We say that X is compactly generated (k -space) if for every set $A \subseteq X$, A is open if and only if $A \cap K$ is open in K for every compact subspace $K \subseteq X$.

Definition 1.1.2 (Category of Compactly Generated Spaces) Define the category of compactly generated spaces $\mathbf{CG}$ to be the full subcategory of $\mathbf{Top}$ consisting of spaces that are compactly generated. In other words, $\mathbf{CG}$ consists of the following data:

- $\text{Obj}(\mathbf{CG})$ consists of all spaces that are compactly generated.
- For $X, Y \in \text{Obj}(\mathbf{CG})$, the morphisms are

$$\text{Hom}_{\mathbf{CG}}(X, Y) = \text{Hom}_{\mathbf{Top}}(X, Y)$$

- Association is given by composition of functions.

Define similarly the category of pointed compactly generated spaces $\mathbf{CG}_*$.

Example 1.1.3

The following are compactly generated spaces.

- Locally compact Hausdorff spaces.
- Topological manifolds.
- Metric spaces.
- CW complexes with finitely many cells in each dimension.

Definition 1.1.4 (New k -space from Old) Let X be a space. Define $k(X)$ to be the set X together with the topology defined as follows: $A \subseteq X$ is open if and only if $A \cap K$ is open in K for every compact subspace $K \subseteq X$.

Lemma 1.1.5 Let X be a space. Then $k(X)$ is a compactly generated space.

Unfortunately $X \times Y$ may not be compactly generated even when X and Y are. But as it turns out, products do exist in $\mathbf{CG}$ and are given by $X \times_{\mathbf{CG}} Y = k(X \times_{\mathbf{Top}} Y)$.

Proposition 1.1.6 Let X, Y be compactly generated spaces. Then the categorical product of X and Y in the category of compactly generated spaces is given by

$$X \times_{\mathbf{CG}} Y = k(X \times_{\mathbf{Top}} Y)$$

Lemma 1.1.7

Let X, Y be spaces. If X is locally compact and Y is compactly generated, then there is a homeomorphism

$$X \times_{\mathbf{CG}} Y \cong X \times_{\mathbf{Top}} Y$$

In particular, since $I = [0, 1]$ is locally compact, $X \times_{\mathbf{CG}} I \cong X \times_{\mathbf{Top}} I$ for a compactly generated spaces X so the notion of homotopy stays the same in this new category.

Definition 1.1.8 (Category of Compactly Generated and Weakly Hausdorff Spaces) Define the category of compactly generated and weakly Hausdorff spaces $\mathbf{CGWH}$ to be the full subcategory of $\mathbf{Top}$ consisting of spaces that are compactly generated and weakly Hausdorff. In other words, $\mathbf{CGWH}$ consists of the following data:

- $\mathbf{Obj}(\mathbf{CGWH})$ consists of all spaces that are compactly generated and weakly Hausdorff.
- For $X, Y \in \mathbf{Obj}(\mathbf{CGWH})$, the morphisms are

$$\mathrm{Hom}_{\mathbf{CGWH}}(X, Y) = \mathrm{Hom}_{\mathbf{Top}}(X, Y)$$

- Association is given by composition of functions.

Define similarly the category of pointed compactly generated spaces $\mathbf{CGWH}_*$.

Proposition 1.1.9 A compactly generated space X is weakly Hausdorff if and only if the diagonal subspace $\Delta = \{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Proposition 1.1.10 Product of $\mathbf{CGWH}$ is $\mathbf{CGWH}$

Proposition 1.1.11

The following are true regarding $\mathbf{CGWH}$.

- $\mathbf{CGWH}$ is complete.
- $\mathbf{CGWH}$ is cocomplete.
- The forgetful functor $\mathbf{CGWH} \rightarrow \mathbf{Sets}$ respects limits.

Proposition 1.1.12

Let $X, Y, Z \in \mathbf{CGWH}$. Then the following are true regarding the mapping space.

- The evaluation map $\mathrm{Map}(X, Y) \times X \rightarrow Y$ is continuous.
- The composition map $\mathrm{Map}(X, Y) \times \mathrm{Map}(Y, Z) \rightarrow \mathrm{Map}(X, Z)$ is continuous.
- The map $\mathrm{Map}(X \times Y, Z) \rightarrow \mathrm{Map}(X, \mathrm{Map}(Y, Z))$ is a homeomorphism.

Theorem 1.1.13 Let $X, Y, Z \in \mathbf{CGWH}$. Then the functors $- \times_{\mathbf{CGWH}} Y : \mathbf{CGWH} \rightarrow \mathbf{CGWH}$ and $\mathrm{Map}(Y, -) : \mathbf{CGWH} \rightarrow \mathbf{CGWH}$ are adjoint functors with the adjunction formula

$$\mathrm{Hom}_{\mathbf{CGWH}}(X \times_{\mathbf{CGWH}} Y, Z) \cong \mathrm{Hom}_{\mathbf{CGWH}}(X, \mathrm{Map}(Y, Z))$$

Moreover, by giving the Hom set the compact open topology and applying k , we obtain an isomorphism

$$\mathrm{Map}(X \times_{\mathbf{CGWH}} Y, Z) \cong \mathrm{Map}(X, \mathrm{Map}(Y, Z))$$

1.2 The Smash Product and the Pointed Mapping Space

Aside from the adjunction between the product space and the mapping space, another major reason one considers compactly generated spaces is that the smash product gives another adjunction.

Definition 1.2.1 (The Smash Product) Let (X, x_0) and (Y, y_0) be pointed topological spaces. Define the smash product of the two pointed spaces to be the pointed space

$$X \wedge Y = \frac{X \times Y}{X \vee Y}$$

together with the point (x_0, y_0) .

Proposition 1.2.2 Let X, Y, Z be compactly generated spaces with a chosen base point. Then the following are true.

- $(X \wedge Y) \wedge Z \cong X \wedge (Y \wedge Z)$
- $X \wedge Y \cong Y \wedge X$

Theorem 1.2.3 The category $\mathbf{CG}$ of compactly generated spaces is a symmetric monoidal category with operator the smash product $\wedge : \mathbf{CG} \times \mathbf{CG} \rightarrow \mathbf{CG}$ and the unit S^0 .

Note that this is not true if we do not restrict the spaces to the category of compactly generated spaces.

Lemma 1.2.4 Let X be a pointed space. Then the reduced suspension and the smash product with the circle

$$\Sigma X \cong X \wedge S^1$$

are homeomorphic spaces.

Theorem 1.2.5 Let X, Y, Z be compactly generated with a chosen basepoint. Then the functors $- \wedge Y : \mathcal{K}_* \rightarrow \mathcal{K}_*$ and $\text{Map}_*(Y, -) : \mathcal{K}_* \rightarrow \mathcal{K}_*$ are adjoint functors with the adjunction formula

$$\text{Hom}_{\mathcal{K}_*}(X \wedge Y, Z) \cong \text{Hom}_{\mathcal{K}_*}(X, \text{Map}_*(Y, Z))$$

Moreover, by giving the Hom set the compact open topology and applying k , we obtain an isomorphism

$$\text{Map}_*(X \wedge Y, Z) \cong \text{Map}_*(X, \text{Map}_*(Y, Z))$$

1.3 The Suspension-Loopspace Adjunction

By choosing $Y = I$ in the adjunction, we recover the usual suspension-loopspace adjunction in $\mathbf{Top}_*$.

Theorem 1.3.1 (The Suspension-Loopspace Adjunction) The functor $\Sigma : \mathbf{Top}_* \rightarrow \mathbf{Top}_*$ is a left adjoint to the functor $\Omega : \mathbf{hTop}_* \rightarrow \mathbf{hTop}_*$. Explicitly, if X, Y are pointed spaces, there is a natural bijection of sets

$$\text{Hom}_{\mathbf{Top}_*}(\Sigma X, Y) \cong \text{Hom}_{\mathbf{Top}_*}(X, \Omega Y)$$

Moreover, the bijection induces a homeomorphism if we equip both sides with the compact open topology.

Theorem 1.3.2 The suspension-loopspace adjunction

$$\Sigma : \mathbf{Top}_* \rightleftarrows \mathbf{Top}_* : \Omega$$

induces an adjunction in the homotopy category

$$\Sigma : \mathbf{hTop}_* \rightleftarrows \mathbf{hTop}_* : \Omega$$

Explicitly, there is a natural bijection given by

$$[\Sigma X, Y]_* \cong [X, \Omega Y]_*$$

Switzer / Hatcher

By the set bijection $[\Sigma^n X, Y]_* \cong [X, \Omega^n Y]_*$, we can endow the structure of a group on $[\Sigma^n X, Y]_*$.

2 Introduction to Fibrations and Cofibrations

2.1 The Homotopy Lifting and Extension Properties

Definition 2.1.1 (The Homotopy Lifting Property)

Let E, B be spaces. Let $p : E \rightarrow B$ be a map. Let X be a space. We say that p has the homotopy lifting property with respect to X if for every homotopy $H : X \times I \rightarrow B$ and a lift $\widetilde{H(-, 0)} : X \rightarrow E$ of $H(-, 0)$, there exists a homotopy $\widetilde{H} : X \times I \rightarrow E$ such that the following diagram commutes:

$$\begin{array}{ccc} X \times \{0\} & \xrightarrow{\widetilde{H(-, 0)}} & E \\ \downarrow \iota & \searrow \exists \widetilde{H} & \downarrow p \\ X \times I & \xrightarrow{H} & B \end{array}$$

Definition 2.1.2 ((Hurewicz) Fibrations)

Let E, B be spaces. Let $p : E \rightarrow B$ be a map. We say that p is a fibration if it has the homotopy lifting property with respect to all topological spaces. In this case, we call B the base space and E the total space.

Definition 2.1.3 (The Homotopy Extension Property) Let $i : A \rightarrow X$ be a map and let Y be a space. Denote i_0 the inclusion map $A \times \{0\} \hookrightarrow A \times I$. We say that i has the homotopy extension property with respect to Y if for every homotopy $H : A \times I \rightarrow Y$ and every map $f : X \rightarrow Y$ such that

$$H \circ i_0 = f \circ i$$

there exists a homotopy $\widetilde{H} : X \times I \rightarrow Y$ such that the following diagram commute:

$$\begin{array}{ccc} A \cong A \times \{0\} & \xrightarrow{i_0} & A \times I \\ \downarrow i & & \downarrow i \times \text{id}_I \\ X \cong X \times \{0\} & \xrightarrow{i_0} & X \times I \end{array} \quad \begin{array}{c} \curvearrowright H \\ \searrow \exists \widetilde{H} \\ \downarrow f \end{array}$$

Definition 2.1.4 (Cofibrations) Let A, X be spaces. Let $i : A \rightarrow X$ be a map. We say that i is a cofibration if it has the homotopy extension property for all spaces Y .

Cofibrations and fibrations are dual in the following sense. Recall from section 1 that if X and Y are in **CGWH**, then there is a bijection

$$\text{Hom}_{\text{CGWH}}(X \times I, Y) \cong \text{Hom}_{\text{CGWH}}(X, \text{Map}(I, Y))$$

Now under this bijection, we can rewrite the diagram in the homotopy lifting property:

$$\begin{array}{ccc} X & \xrightarrow{\quad} & E^I \\ \text{id}_X \downarrow & \searrow \exists \widetilde{H} & \downarrow p_* \\ X & \xrightarrow{H} & B^I \end{array}$$

2.2 Algebra of Fibrations and Cofibrations

Lemma 2.2.1

Let X, Y, Z be spaces. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be maps. Then the following are true.

- If f and g are fibrations, then $g \circ f$ is a fibration.
- If f and g are cofibrations, then $g \circ f$ is a cofibration.

Proof

Apply the homotopy lifting property and the homotopy extension property twice respectively. ■

Proposition 2.2.2

The following are true.

- The product of two fibrations is a fibration.
- The product of two cofibrations is a fibration.
- The pullback of a fibration is a fibration.
- The pushout of a fibration is a fibration.

Proposition 2.2.3 Let X, Y, Z be spaces. Let $f : X \rightarrow Y$ be a map.

- Let f be a fibration. Consider the following lifting problem:

$$\begin{array}{ccc} Z \times \{0\} & \xrightarrow{g} & X \\ i_0 \downarrow & \nearrow & \downarrow f \\ Z \times I & \xrightarrow{h} & Y \end{array}$$

If h_0 and h_1 are both solutions to the lifting problem, then h_0 and h_1 are homotopic relative to $Z \times \{0\}$.

- Let f be a cofibration. Consider the following extension problem:

$$\begin{array}{ccc} X & \xrightarrow{g} & Z \times \{0\} \\ f \downarrow & \nearrow & \downarrow \text{ev}_0 \\ Y & \xrightarrow{h} & Z \times I \end{array}$$

If h_0 and h_1 are both solutions to the extension problem, then h_0 and h_1 are homotopic relative to Z .

Proposition 2.2.4 Let $X, Y, Z \in \mathbf{CGWH}$. Let $p : X \rightarrow Y$ be a map.

- If p is a fibration and $f : Z \rightarrow Y$ is a (homotopy) weak equivalence, then the pullback $X \times_Y Z \rightarrow X$ of p and f is a (homotopy) weak equivalence

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{\simeq} & X \\ \downarrow & & \downarrow p \\ Z & \xrightarrow{f, \simeq} & Y \end{array}$$

- If p is a cofibration and $g : X \rightarrow Z$ is a (homotopy) weak equivalence, then the push forward $Y \rightarrow Z \coprod_X Y$ of p and g is a (homotopy) weak equivalence

$$\begin{array}{ccc} X & \xrightarrow{g, \simeq} & Z \\ p \downarrow & & \downarrow \\ Y & \xrightarrow{\simeq} & Z \coprod_X Y \end{array}$$

Proposition 2.2.5

Let $p : X \rightarrow B$ be a fibration. Let $f, g : X \rightarrow B$ be homotopic. Then $f^*(X)$ and $g^*(X)$ are homotopy equivalent.

Corollary 2.2.6

Let $p : X \rightarrow B$ be a fibration. If B is contractible, then p is homotopy equivalent to the trivial fibration.

Proposition 2.2.7

Let $i : A \rightarrow X$ be a cofibration such that A is closed in X . Let $f, g : A \rightarrow Y$ be homotopic. Then the pushouts of f and g with i are homotopic relative to Y .

Theorem 2.2.8 (The Lifting Axiom)

Suppose that the following diagram lies in **Top**:

$$\begin{array}{ccc} A & \xrightarrow{g} & X \\ i \downarrow & \nearrow f & \downarrow p \\ B & \xrightarrow{h} & Y \end{array}$$

where i is a cofibration and p is a fibration. Then the following are true.

- If i is a homotopy equivalence, then there exists $f : B \rightarrow X$ such that the above diagram commutes.
- If p is a homotopy equivalence, then there exists $f : B \rightarrow X$ such that the above diagram commutes.

2.3 Fibration and Cofibration Replacements

Definition 2.3.1 (The Mapping Path Space)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Define the mapping path space of f to be the set

$$P_f = \{(x, \gamma) \in X \times \text{Map}(I, Y) \mid \gamma(0) = f(x)\}$$

together with the subspace topology of $X \times \text{Map}(I, Y)$.

Mapping path space as a pullback.

Proposition 2.3.2

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then the map

$$q : P_f \rightarrow Y$$

given by $(x, \gamma) \mapsto \gamma(1)$ is a fibration.

Proof

Suppose that we are given a homotopy lifting problem:

$$\begin{array}{ccc} A \times \{0\} & \xrightarrow{g} & P_f \\ \downarrow \iota & \nearrow \tilde{H} & \downarrow q \\ A \times I & \xrightarrow{H} & Y \end{array}$$

We write $g(a) = (g_1(a), g_2(a))$ for the components of g . Now recall that the definition of the mapping path space implies that $f(g_1(a)) = g_2(a)(0)$. By commutativity of the diagram and definition of q we also have $g_2(a)(1) = H(a, 0)$. Define $\tilde{H} : A \times I \rightarrow P_f$ by the formula

$$\tilde{H}(a, t) = (g_1(a), h_2(a, t))$$

where

$$h_2(a, t)(s) = \begin{cases} g_2(a)(1+t)(s) & \text{if } 0 \leq s \leq \frac{1}{1+t} \\ H(a, (1+t)s - 1) & \text{if } \frac{1}{1+t} \leq s \leq 1 \end{cases}$$

The definition of h_2 makes sense because $g_2(a)(1) = H(a, 0)$. By the gluing lemma $h_2(a, t)$ is continuous. h_2 is also continuous in a and t because $g_2(a)$ is a path and g_2 is continuous and H is continuous in both variables and the composite of continuous functions are continuous. Hence $\tilde{H}$ is continuous. Now $\tilde{H}(-, 0) = (g_1(-), g_2(-)) = g(-)$. Thus $\tilde{H}(-, 0)$ is a lift of g . It remains to show that $\tilde{H}$ is a lift of H . We have that

$$\begin{aligned} q(\tilde{H}(a, t)) &= q(g_1(a), h_2(a, t)) \\ &= h_2(a, t)(1) \\ &= H(a, t) \end{aligned}$$

and so we conclude. ■

Proposition 2.3.3

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then there exists a homotopy equivalence $h : X \rightarrow P_f$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \exists h & \nearrow q \\ & P_f & \end{array}$$

where $q : P_f \rightarrow Y$ is the above defined map.

Proof Define the map $h : X \rightarrow P_f$ by $h(x) = (x, e_{f(x)})$. It is easy to see that $q \circ h = f$. I claim that the projection map $p_X : P_f \rightarrow X$ gives the homotopy inverse of h . Define a map $H : P_f \times I \rightarrow P_f$ by

$$H(x, \gamma, t) = (x, \gamma_t)$$

where γ_s is the path $s \mapsto \gamma(st)$. It is continuous since the composition of continuous functions are continuous and each component of H is continuous. Also, we have that $h(p_X(x, \gamma)) = h(x) = (x, e_{f(x)})$ and $H(x, \gamma, 0) = (x, \gamma_0) = (x, e_{f(x)})$ so that $H(-, 0) = h \circ p_X$. When $t = 1$ we also have

$$H(-, 1) = (x, \gamma, 1) = (x, \gamma_1) = (x, \gamma) = \text{id}_{P_f}(x, \gamma)$$

so that H is a homotopy. ■

Definition 2.3.4 (The Mapping Cylinder)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Define the mapping cylinder of f to be the space

$$M_f = \frac{(X \times I) \amalg Y}{(x, 0) \sim f(x)} = (X \times I) \amalg_f Y$$

Mapping cylinder as a push forward.

Proposition 2.3.5

Let A, X be spaces. Let $f : A \rightarrow X$ be a map. Then the map

$$q : A \rightarrow M_f$$

given by $q(a) = [a, 0]$ is a cofibration.

Proof Suppose that we are given a homotopy lifting problem:

$$\begin{array}{ccc}
 A \cong A \times \{0\} & \xrightarrow{\iota_0} & A \times I \\
 \downarrow i & & \downarrow i \times \text{id}_I \\
 X \cong X \times \{0\} & \xrightarrow{\iota_0} & X \times I \\
 & \searrow f & \downarrow \tilde{H} \\
 & & Y
 \end{array}
 \quad
 \begin{array}{c}
 \nearrow H \\
 \downarrow
 \end{array}$$

Proposition 2.3.6

Let A, X be spaces. Let $f : A \rightarrow X$ be a map. Then the inclusion map $i : A \rightarrow M_f$ defined by $i(a) = [a, 0]$ is a cofibration. Moreover, there exists a homotopy equivalence $h : M_f \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc}
 A & \xrightarrow{f} & X \\
 \searrow i & & \nearrow \exists h \\
 & M_f &
 \end{array}$$

2.4 Fibers and Cofibers

Definition 2.4.1 (Fibers of a Map)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Define the fiber of f at $y \in Y$ to be

$$\text{Fib}_y(f) = f^{-1}(y)$$

Definition 2.4.2 (Cofibers of a Map)

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Define the cofiber of f to be

$$\text{Cofib}(f) = \frac{Y}{f(X)}$$

Fibers and Cofibers as pullbacks / pushouts.

Fibers of homotopic fibrations are homotopic. Cofibers of homotopic cofibrations are homotopic.

Proposition 2.4.3

Let X, B be spaces. Let $p : X \rightarrow B$ be a fibration. Let $b_1, b_2 \in B$ lie in the same path component of B . Then there is a homotopy equivalence

$$\text{Fib}_{b_1}(p) \simeq \text{Fib}_{b_2}(p)$$

given by lifting a path from b_1 to b_2 by the homotopy lifting property.

Cofibers in a cofibration are homotopic.

2.5 The Compactly Generated Case

Proposition 2.5.1 Let $X, Y, Z \in \mathbf{CGWH}$. Let $f : X \rightarrow Y$ be a map. Then the following are true.

- If f is a fibration, then the induced map

$$f_* : \mathrm{Map}(Z, X) \rightarrow \mathrm{Map}(Z, Y)$$

is a fibration.

- If f is a cofibration, then the map

$$f \times \mathrm{id}_Z : X \times Z \rightarrow Y \times Z$$

is a cofibration.

3 Topological Properties of Fibrations

3.1 The Topology of Fibrations

Theorem 3.1.1 (Hurewicz)

Let $p : X \rightarrow B$ be a map. Suppose the following are true.

- B is paracompact.
- There exists an open cover $\{U_i \mid i \in I\}$ of B such that $p^{-1}(U_i) \rightarrow B$ is a fibration for all i .

Then p is a fibration.

Lemma 3.1.2

Let $p : X \rightarrow B$ be a fiber bundle. If B is paracompact, then p is a fibration.

Example 3.1.3 (The Hopf Fibration)

Define the Hopf fibration $h : S^3 \rightarrow S^2$ as follows. Consider S^2 as the one point compactification of $\mathbb{C}$. Also consider $S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}$. Define the map h by

$$(z_1, z_2) \mapsto \frac{z_2}{z_1}$$

Then h is a fibration and the fibers are homeomorphic to S^1 .

Proposition 3.1.4 Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $x_0 \in X$. Then the evaluation map

$$\text{ev}_{x_0} : \text{Map}(X, Y) \rightarrow Y$$

defined by $f \mapsto f(x_0)$ is a fibration.

3.2 Fiber Maps and Fiber Homotopy Equivalences

Definition 3.2.1 (Fiber Maps)

Let B, X, Y be spaces. Let $p : X \rightarrow B$ and $q : Y \rightarrow B$ be fiber maps. Let $f : X \rightarrow Y$ be a map. We say that f is a fiber map if $p = q \circ f$.

Definition 3.2.2 (Fiber Homotopy Equivalence)

Let $p : X \rightarrow B$ and $q : Y \rightarrow B$ be fibrations. Let $f : X \rightarrow Y$ be a fiber map. We say that f is a fiber homotopy equivalence if there exists a fiber map $g : Y \rightarrow X$ and homotopies $H : X \times I \rightarrow X$ and $K : Y \times I \rightarrow Y$ such that the following are true.

- H is a homotopy from $g \circ f$ to id_X .
- K is a homotopy from $f \circ g$ to id_Y .
- $p \circ H = p$.
- $q \circ K = q$.

In this case, we say that p and q are fiber homotopy equivalent.

Proposition 3.2.3

Let $p : X \rightarrow B$ and $q : Y \rightarrow B$ be fibrations. Suppose that B is paracompact. Let $f : X \rightarrow Y$ be a map such that $p = q \circ f$. Then the following are true.

- The induced map gives a homotopy equivalence $p^{-1}(b) \simeq q^{-1}(b)$ for all $b \in B$.
- f is a homotopy equivalence.
- f is a fiber homotopy equivalence.

3.3 Serre Fibrations

Definition 3.3.1 (Serre Fibration) We say that a map $p : E \rightarrow B$ is a Serre fibration if it has the homotopy lifting property with respect to all CW-complexes.

Lemma 3.3.2 Every (Hurewicz) fibration is a Serre fibration.

Proof This is true since Hurewicz fibrations satisfies the homotopy lifting property with respect to all topological spaces, including CW complexes. ■

Lemma 3.3.3
Every fiber bundle is a Serre fibration.

3.4 Principal Fibrations

Definition 3.4.1 (Principal Fibrations)

Let E, B be spaces. Let $p : E \rightarrow B$ be a fibration. Let F be the fiber of p at $b \in B$. We say that p is a principal fibration if there exists spaces X, Y, Z and a commutative diagram

$$\begin{array}{ccccccc} F & \longrightarrow & E & \longrightarrow & B \\ \simeq \downarrow & & \downarrow \simeq & & \downarrow \simeq \\ \Omega Z & \longrightarrow & X & \longrightarrow & Y & \longrightarrow & Z \end{array}$$

such that the vertical maps are weak homotopy equivalences and $Y \rightarrow Z$ is a fibration with fiber X .

Proposition 3.4.2

Let F, E, B be spaces. Let $p : \Omega B \rightarrow F \rightarrow E$ be a principal fibration. Then p is fiber homotopy equivalent to the trivial fibration if and only if p admits a section (there exists $s : E \rightarrow F$ such that $p \circ s = \text{id}_E$).

4 Topological Properties of Cofibrations

4.1 The Topology of Cofibrations

Proposition 4.1.1

Let $X, A \in \mathbf{CGWH}$. Let $i : A \rightarrow X$ be a map. Then the following are true.

- If i is a cofibration, then i is an embedding.
- If i is an embedding, then i is a cofibration if and only if $i(A) \hookrightarrow X$ is a cofibration.
- There exists an open subset $i(A) \subseteq U \subseteq X$ such that $i(A)$ is a retract of U .
- $i(A)$ is a closed subset of X .

There is actually an easier way to write out cofibrations when (X, A) is a pair of spaces.

Lemma 4.1.2 Let (X, A) be a pair of spaces with A closed in X . Let $\iota : A \rightarrow X$ be the inclusion. Then ι is a cofibration if and only if for all spaces Y and maps $f : X \rightarrow Y$ and $H : A \times I \rightarrow Y$, there exists a map $\tilde{H} : X \times I \rightarrow Y$ such that the following diagram commutes:

$$\begin{array}{ccc} X \times \{0\} \cup A \times I & \xrightarrow{f \cup H} & Y \\ \downarrow \iota & \nearrow \tilde{H} & \\ X \times I & & \end{array}$$

Example 4.1.3

The following are cofibrations.

- $S^n \hookrightarrow D^{n+1}$.
- $X \rightarrow X \times \{0\} \hookrightarrow X \times I$ for any space $X \in \mathbf{CGWH}$.
- $* \hookrightarrow X$ for any pointed space X .

5 Based Fibrations and Cofibrations

5.1 Non-Degenerate Base Points

Definition 5.1.1 (Well-Pointed Space)

Let (X, x_0) be a pointed space. We say that (X, x_0) is well-pointed if $X \times \{0\} \cup \{x_0\} \times I$ is a retract of $X \times I$. In this case we call the base point non-degenerate.

Proposition 5.1.2 Let (X, x_0) be a pointed space. Then (X, x_0) is well-pointed if and only if the inclusion $x_0 \rightarrow X$ is a cofibration.

Example 5.1.3

Let (X, x_0) be a pointed CW complex. Then (X, x_0) is well pointed.

Proposition 5.1.4

Let (X, x_0) be a pointed space. If (X, x_0) is well-pointed, then the following are true.

- The reduced and unreduced cone are homotopy equivalent.
- The reduced and unreduced suspension are homotopy equivalent.

Proposition 5.1.5

Let X, Y be well pointed spaces. Then there is a homotopy equivalence

$$\Sigma(X \times Y) \simeq \Sigma X \wedge \Sigma Y \wedge \Sigma(X \wedge Y)$$

Definition 5.1.6 (The Whisker Trick)

Let X be a pointed space. Define the whisker of X to be the pointed space

$$W(X) = X \vee (I, 0)$$

together with the base point $1 \in I$.

Lemma 5.1.7

Let X be a pointed space. Then the following are true.

- There are isomorphisms $\pi_n(X) \cong \pi_n(W(X))$ for all $n \in \mathbb{N}$.
- There are isomorphisms $H_n(X) \cong H_n(W(X))$ for all $n \in \mathbb{N}$.

Definition 5.1.8 (The Category of Well-Pointed Spaces)

Define the category of well-pointed spaces $\mathbf{Top}_*^{\text{wp}}$ to be the full subcategory of $\mathbf{Top}_*$ consisting of well-pointed spaces.

Proposition 5.1.9

Let $\iota : \mathbf{Top}_*^{\text{wp}} \hookrightarrow \mathbf{Top}_*$ be the inclusion functor. Then there is an adjunction

$$W : \mathbf{Top}_* \xrightleftharpoons{\quad} \mathbf{Top}_*^{\text{wp}} : \iota$$

Proposition 5.1.10

The above adjunction descends to a natural homeomorphism

$$\text{Map}_*(W(X), Y) \cong \text{Map}_*(X, Y)$$

for any based space X and any well-pointed space Y .

5.2 Based Fibrations**5.3 Based Cofibrations**

6 Homotopy Fibers and Homotopy Cofibers

6.1 Basic Definitions

Definition 6.1.1 (Homotopy Fibers and Cofibers) Let $f : X \rightarrow Y$ be a map. Define the homotopy fiber of f at $y \in Y$ to be

$$\text{hofiber}_y(f) = \{(x, \phi) \in X \times \text{Map}(I, Y) \mid f(x) = \phi(0), \phi(1) = y\} = \text{Fib}_y(P_f \rightarrow Y)$$

Lemma 6.1.2

Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Then we have

$$\text{hofib}_y(f) = \text{Fib}_y(P_f \rightarrow Y)$$

Since the map $P_f \rightarrow Y$ is a fibration, the fibers of $P_f \rightarrow Y$, and hence the homotopy fibers of f are all homotopy equivalent.

Proposition 6.1.3 Let $p : E \rightarrow B$ be a fibration. Then there is a homotopy equivalence

$$\text{Fib}_b(f) \simeq \text{Hofib}_b(f)$$

for each $b \in B$, given by the inclusion map $x \mapsto (x, e_x)$.

Proof Consider the following diagram

$$\begin{array}{ccc} \text{Fib}_b(f) & \xrightarrow{\simeq} & \text{Hofib}_b(f) \\ \downarrow & & \downarrow \\ X & \xrightarrow{h, \simeq} & P_f \\ f \downarrow & & \downarrow q \\ Y & \xrightarrow{\text{id}_Y} & Y \end{array}$$

and apply 2.4.6 to conclude. ■

Proposition 6.1.4 Let X, Y be spaces. Let $f, g : X \rightarrow Y$ be maps. If f, g are homotopic, then for all $y \in Y$, there is a homotopy equivalence

$$\text{hofib}_y(f) \simeq \text{hofib}_y(g)$$

induced by the homotopy from f to g .

Example 6.1.5 Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map.

- The homotopy fiber of $\{y\} \hookrightarrow Y$ is given by $\text{hofib}_y(\{y\} \hookrightarrow Y) \cong \Omega Y$
- If f is null-homotopic, then the homotopy fiber of f is given by $\text{hofib}_y(f) \cong X \times \Omega Y$

Example 6.1.6

There is a homotopy equivalence

$$\text{hofib}_z(X \vee Y \hookrightarrow X \times Y) \simeq \Omega X * \Omega Y$$

for any $z \in X \times Y$.

Definition 6.1.7 (Homotopy Fibers and Cofibers) Let $f : X \rightarrow Y$ be a map. Define the homo-

topy cofiber of f to be

$$\text{hocofiber}(f) = \frac{(X \times I) \amalg Y}{(x, 1) \sim f(x), (x, 0) \sim (x', 0)} = \text{Cofib}(X \rightarrow M_f) = C_f$$

Proposition 6.1.8 Let $f : X \rightarrow Y$ be a cofibration. Then there is a homotopy equivalence

$$\text{hocofib}(f) \simeq \text{cofib}(f)$$

given by $(x, t) \mapsto (x, 1)$.

Proposition 6.1.9 Let X, Y be spaces. Let $f, g : X \rightarrow Y$ be maps. If f, g are homotopic, then for all $y \in Y$, there is a homotopy equivalence

$$\text{hocofib}(f) \simeq \text{hocofib}(g)$$

induced by the homotopy from f to g .

6.2 Homotopy Fiber and Cofiber Sequences

Definition 6.2.1 (Homotopy Fiber Sequences)

Let X, Y, Z be spaces. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be maps. We say that the diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

is a homotopy fiber sequence if the following are true.

- $g \circ f$ is homotopic to the constant map to some $z \in Z$.
- The induced map $X \rightarrow \text{hofib}_z(f)$ is a weak homotopy equivalence.

Realistically there are only two examples to keep in mind.

Example 6.2.2

The following are homotopy fiber sequences.

- The diagram $\text{hofib}_y(f) \rightarrow X \xrightarrow{f} Y$ for any $y \in Y$.
- The diagram $\text{Fib}_y(f) \rightarrow X \xrightarrow{f} Y$ if $y \in Y$ and f is a fibration.

Theorem 6.2.3 (The Long Homotopy Fiber Sequence) Let X, Y be pointed spaces. Let $f : X \rightarrow Y$ be a map. Let $F_f = \text{hofib}_y(f)$ denote the homotopy fiber. Define the following maps.

- The map $-\Omega f : \Omega X \rightarrow \Omega Y$ given by $\gamma \mapsto \overline{f \circ \gamma}$ (reversing the traversal)
- The map $k : \Omega Y \hookrightarrow F_f$ given by inclusion.

Consider the following diagram of spaces:

$$\cdots \longrightarrow \Omega^2 X \xrightarrow{\Omega^2 f} \Omega^2 Y \xrightarrow{-\Omega k} \Omega F_f \xrightarrow{-\Omega \text{incl}} \Omega X \xrightarrow{-\Omega f} \Omega Y \xrightarrow{k} F_f \xrightarrow{\text{incl}} X \xrightarrow{f} Y$$

Then up to homotopy equivalence, every two consecutive maps form a fiber sequence.

Definition 6.2.4 (Exact Sequence of Sets) Let A, B, C be pointed sets. Let $f : A \rightarrow B$ and $g : B \rightarrow C$. We say that the diagram

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact if $\text{im}(f) = g^{-1}(c_0)$ where c_0 is the base point of C .

Theorem 6.2.5 (The Homotopy Fiber Long Exact Sequence) Let X, Y be pointed spaces. Let $f : X \rightarrow Y$ be a map. Let $F_f = \text{hofiber}_y(f)$ denote the homotopy fiber. Then for any space Z ,

$$\cdots \longrightarrow [Z, \Omega^2 X] \xrightarrow{\Omega^2 f \circ -} [Z, \Omega^2 Y] \xrightarrow{-\Omega k \circ -} [Z, \Omega F_f] \xrightarrow{-\Omega \text{incl} \circ -} [Z, \Omega X] \xrightarrow{-\Omega f \circ -} [Z, \Omega Y]$$

is an exact sequence of abelian groups. Moreover, the diagram of sets

$$[Z, \Omega Y] \xrightarrow{k \circ -} [Z, F_f] \xrightarrow{\text{incl} \circ -} [Z, X] \xrightarrow{f \circ -} [Z, Y]$$

is an exact sequence of sets.

Definition 6.2.6 (Homotopy Cofiber Sequence)

Let X, Y, Z be spaces. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be maps. We say that the diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

is a homotopy cofiber sequence if the following are true.

- $g \circ f$ is homotopic to the constant map.
- The induced map $\text{hocofib}(f) \rightarrow Z$ is a weak homotopy equivalence.

Theorem 6.2.7 Let X, Y be pointed spaces. Let $f : X \rightarrow Y$ be a map. Define the following maps.

- The projection map $\pi : C_f \rightarrow \frac{C_f}{Y} \cong \Sigma X$.
- The map $-\Sigma f : \Sigma X \rightarrow \Sigma Y$ given by $x \wedge t \mapsto f(x) \wedge (1 - t)$.

Consider the following diagram of spaces:

$$X \xrightarrow{f} Y \xrightarrow{\text{incl}} C_f \xrightarrow{\pi} \Sigma X \xrightarrow{-\Sigma f} \Sigma Y \xrightarrow{-\Sigma \text{incl}} \Sigma C_f \xrightarrow{-\Sigma \pi} \Sigma^2 X \xrightarrow{\Sigma^2 f} \Sigma^2 Y \longrightarrow \cdots$$

Then up to homotopy equivalence, every two consecutive maps form a cofiber sequence.

Theorem 6.2.8 Let X, Y be pointed spaces. Let $f : X \rightarrow Y$ be a map. Then for any space Z ,

$$\cdots \longrightarrow [\Sigma^2 Y, Z] \xrightarrow{-\circ \Sigma^2 f} [\Sigma^2 X, Z] \xrightarrow{-\circ (-\Sigma \pi)} [\Sigma C_f, Z] \xrightarrow{-\circ (-\Sigma \text{incl})} [\Sigma Y, Z] \xrightarrow{-\circ (-\Sigma f)} [\Sigma X, Z]$$

is an exact sequence of abelian groups. Moreover, the diagram of sets

$$[\Sigma X, Z] \xrightarrow{-\circ \pi} [C_f, Z] \xrightarrow{-\circ \text{incl}} [Y, Z] \xrightarrow{-\circ f} [X, Z]$$

is an exact sequence of sets.

Theorem 6.2.9 (Homotopy Long Exact Sequence in Fibration)

Let (B, b_0) be a pointed path connected space. Let $p : (E, e_0) \rightarrow (B, b_0)$ be a fibration with fiber F . Let $\iota : F \hookrightarrow E$ be the inclusion of the fiber. Then there is a long exact sequence in homotopy groups:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \pi_n(F, e_0) & \xrightarrow{\iota_*} & \pi_n(E, e_0) & \xrightarrow{p_*} & \pi_n(B, b_0) \longrightarrow \pi_{n-1}(F, e_0) \\ & & & & & & \searrow \\ & & & & & & \pi_0(F, e_0) \xrightarrow{\iota_*} \pi_0(E, e_0) \longrightarrow 0 \end{array}$$

for $e_0 \in E$ and $b_0 = p(e_0)$. Moreover, p_* is an isomorphism.

We demonstrate the power of this theorem with a few computation of the homotopy groups of spheres. Recall that the Hopf fibrations are the fiber bundles of the form

- $S^1 \hookrightarrow S^3 \rightarrow \mathbb{CP}^1 \cong S^2$.

- $S^3 \hookrightarrow S^7 \rightarrow \mathbb{H}\mathbb{P}^1 \cong S^4$.
- $S^7 \hookrightarrow S^{15} \rightarrow \mathbb{O}\mathbb{P}^1 \cong S^8$.

Example 6.2.10

The following are true regarding the homotopy groups of spheres.

- $\pi_n(S^3) \cong \pi_n(S^2)$ for all $n \geq 3$.
- $\pi_n(S^4) \cong \pi_n(S^7) \times \pi_{n-1}(S^3)$ for all $n \geq 2$.
- $\pi_n(S^8) \cong \pi_n(S^{15}) \times \pi_{n-1}(S^7)$ for all $n \geq 2$.

Example 6.2.11

$\mathbb{C}\mathbb{P}^\infty$ is a $K(\mathbb{Z}, 2)$ space.

Proof

From Algebraic Topology 3, we know that S^∞ is weakly contractible. By the long exact sequence of fibration applied to the fiber bundle $S^1 \hookrightarrow S^\infty \rightarrow \mathbb{C}\mathbb{P}^\infty$, we deduce that the only non-trivial homotopy groups are $\pi_2(\mathbb{C}\mathbb{P}^\infty) \cong \pi_1(S^1) \cong \mathbb{Z}$. ■

Example 6.2.12

Let $f : S^n \rightarrow S^n$ be a homeomorphism of degree -1 . Recall that the mapping torus $p : T_f = \frac{S^n \times I}{(x,0) \sim (f(x),1)} \rightarrow S^1$ is a fiber bundle and hence a fibration. Suppose that $n \geq 2$. By the homotopy long exact sequence in fibration, we deduce that

$$\pi_k(T_f) = \begin{cases} \mathbb{Z} & \text{if } k = 1 \\ \pi_k(S^n) & \text{otherwise} \end{cases}$$

While the theory of covering spaces provided great insight for the structure of the fundamental group as well the space itself, the theory no longer works for higher homotopy groups because of the long exact sequence.

Corollary 6.2.13 Let (X, x_0) be a pointed space and let $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space. Then p induces isomorphisms

$$\pi_n(p) : \pi_n(\tilde{X}, \tilde{x}_0) \xrightarrow{\cong} \pi_n(X, x_0)$$

for all $n \geq 2$.

Proof

By the long exact sequence of fibration, any covering map $p : \tilde{X} \rightarrow X$ have discrete fiber and have 0 homotopy groups. This induces isomorphisms when $n \geq 2$. ■

When $n = 1$, we are left with an exact sequence of the form

$$0 \rightarrow \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0) \rightarrow \pi_0(F, \tilde{x}_0) \rightarrow 0$$

which matches covering space theory saying that the homotopy group of a covering space is a subgroup of $\pi_1(X, x_0)$. If the covering space is simply connected (the universal cover), this short exact sequence also shows that $\pi_1(X, x_0)$ is bijective as a set to the path connected components of F , which is just F since the fibers of a covering space is discrete.

Example 6.2.14

$\mathbb{R}\mathbb{P}^\infty$ is a $K(\mathbb{Z}/2\mathbb{Z}, 1)$ space.

Proof

Since each $\mathbb{R}\mathbb{P}^n$ is a path connected, we know that the direct limit $\mathbb{R}\mathbb{P}^\infty$ is also path connected. For each $n \in \mathbb{N}$, we know that S^n is a double cover of $\mathbb{R}\mathbb{P}^n$. By definition of the colimit, we know that S^∞ is also a double cover of $\mathbb{R}\mathbb{P}^\infty$. Since S^∞ is contractible, by the above corollary we conclude

that $\pi_n(\mathbb{RP}^\infty) = 0$ for all $n \geq 2$. Finally, since S^∞ is a universal double cover of $\mathbb{RP}^\infty$, by covering space theory we conclude that $\pi_1(\mathbb{RP}^\infty) \cong \mathbb{Z}/2\mathbb{Z}$. ■

Theorem 6.2.15 (Homology Long Exact Sequence in Cofibration) Let $p : X \rightarrow Y$ be a cofibration with cofiber $C = \frac{Y}{p(X)}$. Let $\text{proj} : Y \rightarrow C$ be the projection map. Then there is a long exact sequence in homology groups:

$$\cdots \longrightarrow \tilde{H}_{n+1}(C) \xrightarrow{\partial} \tilde{H}_n(X) \xrightarrow{f_*} \tilde{H}_n(Y) \xrightarrow{\text{proj}_*} \tilde{H}_n(C) \xrightarrow{\partial} \tilde{H}_{n-1}(X) \longrightarrow \cdots \longrightarrow \tilde{H}_0(Y) \xrightarrow{\text{proj}_*} \tilde{H}_0(B, b_0)$$

6.3 n-Connected Maps

Definition 6.3.1 (n-Connected Maps) Let X, Y be pointed spaces. Let $f : X \rightarrow Y$ be a map. We say that f is n -connected if the induced map

$$\pi_k(f) : \pi_k(X) \rightarrow \pi_k(Y)$$

is an isomorphism for $0 \leq k < n$ and a surjection for $k = n$.

We can rephrase some of the corner stone theorems of homotopy theory using n -connected maps.

- The homotopy excision theorem can be rephrased into the following. For X a CW-complex and A, B sub complexes of X such that $X = A \cup B$ and $A \cap B \neq \emptyset$. If $(A, A \cap B)$ is m -connected and $(B, A \cap B)$ is n -connected for $m, n \geq 0$, then the inclusion

$$\iota : (A, A \cap B) \rightarrow (X, B)$$

is $(m + n)$ -connected.

- The Freudenthal suspension theorem says that if X is an n -connected CW complex, then the map

$$\Omega\Sigma : X \rightarrow \Omega(\Sigma(X))$$

is a $(2n + 1)$ -connected map.

Proposition 6.3.2 Let X, Y be pointed spaces. Let $f : X \rightarrow Y$ be a map. Then the following are equivalent.

- f is k -connected.
- $\text{Hofib}_y(f)$ is $(k - 1)$ -connected for all $y \in Y$.
- (M_f, X) is a k -connected space.

Lemma 6.3.3 Let (X, x_0) be a pointed space. Then (X, x_0) is n -connected if and only if the inclusion map $\iota : \{x_0\} \hookrightarrow X$ is n -connected.

Proposition 6.3.4 Let X be a space. Let $n \in \mathbb{N}$. Then the following are equivalent.

- X is n -connected.
- $\pi_k(X) = 0$ for all $0 \leq k \leq n$.
- For all $-1 \leq k \leq n$, every map $S^k \rightarrow X$ extends to a map $D^{k+1} \rightarrow X$.
- (CX, X) is $(n + 1)$ -connected.

Proposition 6.3.5 Let (X, A) be a space. Let $n \in \mathbb{N}$. Then the following are equivalent.

- (X, A) is n -connected
- For all $0 < k \leq n$, $\pi_k(X, A) = 0$ and $\pi_0(A) \rightarrow \pi_0(X)$ is surjective.
- $\iota : A \hookrightarrow X$ is n -connected.

Proposition 6.3.6 Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let $n \in \mathbb{N}$. If X is not empty, then the following are equivalent.

- f is n -connected
- $\text{hofiber}_y(f)$ is n -connected for all $y \in Y$.
- For all $0 < k \leq n$, $\pi_k(f)$ is an isomorphism and $\pi_n(f)$ is surjective.

Proposition 6.3.7 Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. If f is k -connected, then $\text{hocofib}(f)$ is k -connected.

Proposition 6.3.8 Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Let X be simply connected. Then $\text{hocofib}(f)$ is k -connected if and only if f is k -connected.

7 Homotopy Pullbacks and Pushouts

Pushouts and pullbacks are not well suited for studying homotopy invariant properties. Most of the invariants in algebraic topology detect differences up to homotopy equivalence, but pushouts and pullbacks are more rigid than homotopy equivalence:

- Their universal properties guarantee that they are unique up to homeomorphism.
- Pushouts and pullbacks are determined by three spaces and two maps. But if we supply homotopy equivalent spaces then the pushout / pullback is not homotopy equivalence.

The following example is given in [?].

Example 7.0.1 Consider the following commutative diagram:

$$\begin{array}{ccccc} * & \longleftarrow & S^n & \hookrightarrow & D^{n+1} \\ \downarrow & & \text{id} \downarrow & & \downarrow * \\ * & \longleftarrow & S^n & \longrightarrow & * \end{array}$$

While all the vertical arrows are weak equivalences, the induced map of pushouts is given by $S^{n+1} \rightarrow *$ which is clearly not a weak equivalence.

Therefore we concern ourselves with a homotopy invariant version of this concept. The ordinary pushouts and pullbacks are unique up to homeomorphism. We can explicitly define a set and its topology and show that it satisfies a universal property. We take a similar approach here and first introduce a model for homotopy pushouts. Our main reference for this section is [?].

7.1 The Standard Models for Homotopy Pushouts and Pullbacks

Definition 7.1.1 (Standard Homotopy Pushout)

Let $W, X, Y \in \mathbf{Top}$ be spaces. Let $f : W \rightarrow X$ and $g : W \rightarrow Y$ be maps. Define the standard homotopy pushout of f and g to be the quotient space

$$\text{hocolim}(X \xleftarrow{f} W \xrightarrow{g} Y) = \frac{X \amalg (W \times I) \amalg Y}{\sim}$$

where $\sim$ is the equivalence relation generated by $f(w) \sim (w, 0)$ and $g(w) \sim (w, 1)$ for $w \in W$.

Definition 7.1.2 (Standard Homotopy Pullback)

Let $X, Y, Z \in \mathbf{Top}$ be spaces. Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be maps. Define the standard homotopy pullback of f and g to be the subspace

$$\text{holim}(X \xrightarrow{f} Z \xleftarrow{g} Y) = \{(x, \alpha, y) \in X \times \text{Map}(I, Z) \times Y \mid \alpha(0) = f(x), \alpha(1) = g(y)\}$$

We will use the following two ways to recognize homotopy pushouts and homotopy pullbacks as certain limits and colimits.

Proposition 7.1.3

Let $X, Y, Z \in \mathbf{Top}$ be spaces. Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be maps. Then there is a homeomorphism

$$\lim(X \xrightarrow{f} Z \leftarrow P_g) \cong \text{holim}(X \xrightarrow{f} Z \xleftarrow{g} Y)$$

given by the map $(x, (y, \gamma)) \mapsto (x, \gamma, y)$.

Proof Recall that the limit on the left is given by $\{(x, (y, \gamma)) \in X \times P_g \mid \gamma(1) = f(x)\}$. Also $(y, \gamma) \in P_g$ means that $\gamma(0) = g(y)$. So the map defined by $(x, (y, \gamma)) \mapsto (x, \gamma, y)$ is a well defined map and is clearly a homeomorphism. [?] ■

Notice that we could have equally replaced the space X with P_f to obtain a homeomorphism between

the limit and homotopy pullback entirely by symmetry.

Dually we have a homeomorphism

$$\operatorname{colim}(X \xleftarrow{f} Z \rightarrow M_g) \cong \operatorname{hocolim}(X \xleftarrow{f} Z \xrightarrow{g} Y)$$

Moreover, since the diagram of spaces is symmetric, we could have equally replaced the space on the left by the mapping path space / mapping cylinder and we would yet again obtain a homeomorphism.

Proposition 7.1.4

Let $X, Y, Z \in \mathbf{Top}$ be spaces. Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be maps. Then there is a homeomorphism

$$\lim(P_f \rightarrow Z \leftarrow P_g) \cong \operatorname{holim}(X \xrightarrow{f} Z \xleftarrow{g} Y)$$

given by the map $((x, \gamma_1), (y, \gamma_2)) \mapsto (x, \gamma_1 \cdot \overline{\gamma_2}, y)$.

Proof Notice that the map is well defined because by definition, we have that

$$\lim(P_f \rightarrow Z \leftarrow P_g) = \{((x, \gamma_1), (y, \gamma_2)) \in P_f \times P_g \mid \gamma_1(0) = f(x), \gamma_2(0) = g(y), \gamma_1(1) = \gamma_2(1)\}$$

The evident inverse of the given map is then $(x, \gamma, y) \mapsto ((x, \gamma|_{[0,1/2]}), (y, \gamma_{[1/2,1]}))$. [?] ■

Dually we have a homeomorphism

$$\operatorname{colim}(M_f \leftarrow W \rightarrow M_g) \cong \operatorname{hocolim}(X \xleftarrow{f} W \xrightarrow{g} Y)$$

Proposition 7.1.5

Let X, Y, Z be spaces. Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be maps. If one of f or g is a fibration, then the canonical map

$$\lim(X \rightarrow Z \leftarrow Y) \simeq \operatorname{holim}(X \rightarrow Z \leftarrow Y)$$

is a homotopy equivalence.

Proof Without loss of generality suppose that f is a fibration. Replace g by a homotopy equivalence $h : Y \rightarrow P_g$ and a fibration $k : P_g \rightarrow Z$. We then obtain a commutative diagram:

$$\begin{array}{ccc} \lim(X \rightarrow Z \leftarrow Y) & \longrightarrow & Y \\ \downarrow & & \downarrow h \\ \operatorname{holim}(X \rightarrow Z \leftarrow Y) & \longrightarrow & P_g \\ \downarrow & & \downarrow k \\ X & \xrightarrow{f} & Z \end{array}$$

It is commutative because the map from the limit to X is precisely the projection map to X . Since the outer rectangle and the bottom square are pullbacks, the pasting law ?? shows that the upper square is also a pullback. Now since f is a fibration and the bottom square is a pullback, we conclude that the middle horizontal arrow is a fibration by ??. Since h is a homotopy equivalence, we conclude that the map from the limit to the homotopy limit is a homotopy equivalence by ??. Hence we conclude. ■

We can compute some examples of homotopy pushouts and pullbacks.

Example 7.1.6

Let $f : X \rightarrow Y$ be a map. For any $y \in Y$ recall that the homotopy fiber of f is defined to be the space

$$\operatorname{hofib}_y(f) = \{(x, \gamma) \in X \times \operatorname{Map}(I, Y) \mid \gamma(0) = f(x), \gamma(1) = y\}$$

This is precisely the definition of $\operatorname{holim}(X \xrightarrow{f} Y \leftarrow \{y\})$. Also note that if we take actual limits instead of homotopy limits then the same diagram would give us the fiber of f , thus justifying its

name. By replacing the space X with P_f , we a homeomorphism

$$\mathrm{hofib}_y(X \rightarrow Y \leftarrow \{y\}) \cong \lim(P_f \rightarrow Y \leftarrow \{y\}) = \mathrm{fib}_y(P_f \rightarrow Y)$$

Similarly, the homotopy cofiber of f is defined to be

$$\mathrm{hocofib}(f) = \frac{(X \times I) \amalg Y}{\sim}$$

where the relation is generated by $(x, 1) \sim (x', 1)$ and $(x, 0) \sim f(x)$ for $x, x' \in X$. This is precisely the definition of $\mathrm{hocolim}(* \leftarrow X \xrightarrow{f} Y)$.

Example 7.1.7

Let X be a pointed space. There is a unique map $X \rightarrow *$ to the terminal object in $\mathbf{Top}$. The homotopy pushout of $* \leftarrow X \rightarrow *$ is given by

$$\mathrm{hocolim}(* \leftarrow X \rightarrow *) = \frac{* \amalg (X \times I) \amalg *}{\sim}$$

where $\sim$ is generated by $* \sim (x, 0)$ and $* \sim (x, 1)$. This is precisely the definition of ΣX . Similarly, there is a unique map $* \rightarrow X$ sending $*$ to the base point of X . The homotopy pullback of $* \rightarrow X \leftarrow *$ is given by

$$\begin{aligned} \mathrm{holim}(* \rightarrow X \leftarrow *) &= \{(*, \gamma, *) \in * \times \mathrm{Map}(I, X) \times * \mid \gamma(0) = * = \gamma(1)\} \\ &\cong \{\gamma \in \mathrm{Map}(I, X) \mid \gamma \text{ is a loop at the base point}\} \end{aligned}$$

This is precisely the definition of ΩX .

Example 7.1.8

Let $f : X \rightarrow Y$ be a map. The homotopy pullback of f and the identity map on Y is given by

$$\begin{aligned} \mathrm{holim}(X \xrightarrow{f} Y \xleftarrow{\mathrm{id}} Y) &= \{(x, \gamma, y) \in X \times \mathrm{Map}(I, Y) \times Y \mid \gamma(0) = x, \gamma(1) = y\} \\ &\cong \{(x, \gamma) \in X \times \mathrm{Map}(I, Y) \mid \gamma(0) = x\} \\ &= P_f \end{aligned}$$

In particular, the homotopy pullback is homotopy equivalent to X by ??

Similarly, the homotopy pushout of f and the identity map on X is given by

$$\begin{aligned} \mathrm{hocolim}(Y \xleftarrow{f} X \xrightarrow{\mathrm{id}} X) &= \frac{Y \amalg (X \times I) \amalg X}{f(x) \sim (x, 0), x \sim (x, 1)} \\ &\cong \frac{Y \amalg (X \times I)}{f(x) \sim (x, 0)} \\ &= M_f \end{aligned}$$

In particular, the homotopy pushout is homotopy equivalent to Y .

Example 7.1.9

Let X, Y be pointed spaces. The homotopy pushout of the initial maps $* \rightarrow X$ and $* \rightarrow Y$ that maps the unique object to the base point is given by

$$\mathrm{hocolim}(X \leftarrow * \rightarrow Y) \cong \frac{X \amalg I \amalg Y}{x_0 \sim 0, y_0 \sim 1}$$

which is just the spaces X and Y together with the unit interval connecting the base points. Since I deformation retracts to a point, we see that the homotopy colimit is homotopy equivalent to $X \vee Y$. Another way to see this is that because the two initial maps are cofibration, the homotopy pushout is homotopy equivalent to the ordinary limit.

Dually, the homotopy pullback of the final maps $X \rightarrow *$ and $Y \rightarrow *$ is given by

$$\begin{aligned} \operatorname{holim}(X \rightarrow * \leftarrow Y) &= \{(x, \gamma, y) \in X \times \operatorname{Map}(I, *) \times Y \mid \gamma(0) = * = \gamma(1)\} \\ &\cong X \times \operatorname{Map}(I, *) \times Y \cong X \times Y \end{aligned}$$

Example 7.1.10

Let X, Y be spaces. There is an evident inclusion map $X \vee Y \rightarrow X \times Y$. We can ask for its homotopy pushout with the terminal map $X \vee Y \rightarrow *$. But $X \vee Y \rightarrow X \times Y$ is a closed inclusion and so is a cofibration (Refer to section 2.4 of [?]). Therefore the homotopy pushout is precisely the cofiber of this map, which is the quotient $X \wedge Y = \frac{X \times Y}{X \vee Y}$.

Proposition 7.1.11

Suppose that the following is a commutative diagram of spaces:

$$\begin{array}{ccccc} X_1 & \xrightarrow{h_1} & X_3 & \xleftarrow{h_2} & X_2 \\ f_1 \downarrow & & f_3 \downarrow & & \downarrow f_2 \\ Z_1 & \xrightarrow{l_1} & Z_3 & \xleftarrow{l_2} & Z_2 \\ g_1 \uparrow & & g_3 \uparrow & & \uparrow g_2 \\ Y_1 & \xrightarrow{k_1} & Y_3 & \xleftarrow{k_2} & Y_2 \end{array}$$

Let R_i be the homotopy pullback of the i th row, let C_i be the homotopy pullback of the i th column. Then there is a homeomorphism

$$\operatorname{holim}(R_1 \rightarrow R_3 \leftarrow R_2) \cong \operatorname{holim}(C_1 \rightarrow C_3 \leftarrow C_2)$$

Proof Using the definition of the homotopy pullback we have

$$\operatorname{holim}(C_1 \rightarrow C_3 \leftarrow C_2) = \left\{ ((x_1, \alpha_1, y_1), \Gamma, (x_2, \alpha_2, y_2)) \in C_1 \times \operatorname{Map}(I, C_3) \times C_2 \mid \begin{array}{l} \Gamma(0) = (f_1(x_1), l_1 \circ \alpha_1, g_1(y_1)) \\ \Gamma(1) = (f_2(x_2), l_2 \circ \alpha_2, g_2(y_2)) \end{array} \right\}$$

Notice that the data of Γ is equivalent to a homotopy $\Gamma : I \times I \rightarrow Z_3$, from $\Gamma(0, -) : l_1 \circ \alpha_1$ to $\Gamma(1, -) : l_2 \circ \alpha_2$. Similarly, we have

$$\operatorname{holim}(R_1 \rightarrow R_3 \leftarrow R_2) = \left\{ ((x_1, \beta_X, x_2), \Gamma, (y_1, \beta_Y, y_2)) \in R_1 \times \operatorname{Map}(I, R_3) \times R_2 \mid \begin{array}{l} \Gamma(0) = (h_1(x_1), f_3 \circ \beta_X, h_2(x_2)) \\ \Gamma(1) = (k_1(y_1), g_3 \circ \beta_Y, k_2(y_2)) \end{array} \right\}$$

and again the data of Γ is equivalent to a homotopy $I \times I \rightarrow Z_3$, from $f_3 \circ \beta_X$ to $g_3 \circ \beta_Y$. Then the map $\operatorname{holim}(C_1 \rightarrow C_3 \leftarrow C_2) \rightarrow \operatorname{holim}(R_1 \rightarrow R_3 \leftarrow R_2)$ defined by

$$((x_1, \alpha_1, y_1), \Gamma, (x_2, \alpha_2, y_2)) \mapsto ((x_1, \Gamma(-, 0), x_2), \Gamma, (y_1, \Gamma(-, 1), y_2))$$

gives the desired homeomorphism. ■

The dual of the above is also true. Namely, if we have a commutative diagram of spaces:

$$\begin{array}{ccccc} X_1 & \longleftarrow & X_3 & \longrightarrow & X_2 \\ \uparrow & & \uparrow & & \uparrow \\ Z_1 & \longleftarrow & Z_3 & \longrightarrow & Z_2 \\ \downarrow & & \downarrow & & \downarrow \\ Y_1 & \longleftarrow & Y_3 & \longrightarrow & Y_2 \end{array}$$

Then the homotopy pushout of the homotopy pushout of the rows is homeomorphic to the homotopy pushout of the homotopy pushout of the columns. Using this we can show the following:

Example 7.1.12

Let X, Y, Z, W be pointed spaces. Let $f : X \rightarrow Y$ and $g : X \rightarrow Z$ be maps. Then there is a homeomorphism

$$W \wedge \operatorname{hocolim}(Y \leftarrow X \rightarrow Z) \cong \operatorname{holim}(W \wedge Y \leftarrow W \wedge X \rightarrow W \wedge Z)$$

To prove this, notice that there is a homeomorphism

$$\operatorname{holim}(W \times Y \leftarrow W \times X \rightarrow W \times Z) \cong W \times \operatorname{hocolim}(Y \leftarrow X \rightarrow Z)$$

given by the identity map.

We can also show that wedge sums commute with homotopy pushouts. To see this, consider the following diagram:

$$\begin{array}{ccccc} W & \longleftarrow & W & \longrightarrow & W \\ \uparrow & & \uparrow & & \uparrow \\ * & \longleftarrow & * & \longrightarrow & * \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & X & \longrightarrow & Z \end{array}$$

For each column, its homotopy pushout is precisely the wedge of W and the corresponding space because the initial map $* \rightarrow W$ is a cofibration. Therefore the homotopy pushout of the homotopy pushout is the homotopy pushout of wedge sums. On the other hand, the homotopy pushout of the first row is W , the homotopy pushout of the second row is $*$ hence the homotopy pushout of the homotopy pushout of the rows is given by $W \vee \operatorname{hocolim}(Y \leftarrow X \rightarrow Z)$.

Then consider the commutative diagram:

$$\begin{array}{ccccc} W \times Y & \longleftarrow & W \times X & \longrightarrow & W \times Z \\ \uparrow & & \uparrow & & \uparrow \\ W \vee Y & \longleftarrow & W \vee X & \longrightarrow & W \vee Z \\ \downarrow & & \downarrow & & \downarrow \\ * & \longleftarrow & * & \longrightarrow & * \end{array}$$

where the top vertical arrows are inclusions. The homotopy pushout of each column is precisely the corresponding smash product by W . The homotopy pushout of the homotopy pushout of the rows is given by

$$\operatorname{hocolim}(W \times \operatorname{hocolim}(Y \leftarrow X \rightarrow Z) \leftarrow W \vee \operatorname{hocolim}(Y \leftarrow X \rightarrow Z) \rightarrow *)$$

Therefore we have homeomorphisms

$$\begin{aligned} W \wedge \operatorname{hocolim}(Y \leftarrow X \rightarrow Z) &\cong \operatorname{hocolim}(W \times \operatorname{hocolim}(Y \leftarrow X \rightarrow Z) \leftarrow W \vee \operatorname{hocolim}(Y \leftarrow X \rightarrow Z) \rightarrow *) \\ &\cong \operatorname{hocolim}(W \wedge Y \leftarrow W \wedge X \rightarrow W \wedge Z) \end{aligned}$$

By taking $W = S^1$, we also deduce that suspension commutes with taking standard homotopy pushouts.

Recall that ordinary pullbacks are not well-suited for studying homotopy invariant concepts. The following theorem shows that homotopy pullbacks remedies the situation.

Proposition 7.1.13

Suppose that we have a commutative diagram of spaces

$$\begin{array}{ccccc} X & \xrightarrow{f} & Z & \xleftarrow{g} & Y \\ e_X \downarrow & & e_Z \downarrow & & \downarrow e_Y \\ X' & \xrightarrow{f'} & Z' & \xleftarrow{g'} & Y' \end{array}$$

in **Top**. Define the map

$$\phi_{X,Z,Y}^{X',Z',Y'} : \text{holim}(X \xrightarrow{f} Z \xleftarrow{g} Y) \rightarrow \text{holim}(X' \xrightarrow{f'} Z' \xleftarrow{g'} Y')$$

by the formula $(x, \gamma, y) \mapsto (e_X(x), e_Z \circ \gamma, e_Y(y))$. If each e_X, e_Y, e_Z are homotopy equivalences, then ϕ is a homotopy equivalence.

Proof Consider the following commutative diagram:

$$\begin{array}{ccccc} X & \xrightarrow{f} & Z & \xleftarrow{g} & Y \\ \text{id}_X \downarrow & & \downarrow e_Z & & \downarrow \text{id}_Y \\ X & \xrightarrow{e_Z \circ f} & Z' & \xleftarrow{e_Z \circ g} & Y \\ e_X \downarrow & & \downarrow \text{id}_{Z'} & & \downarrow e_Y \\ X' & \xrightarrow{f'} & Z' & \xleftarrow{g'} & Y' \end{array}$$

We prove that the homotopy pullback of the first row is homotopy equivalent to that of the second, and we prove that the homotopy pullback of the second row is homotopy equivalent to that of the third.

Since e_Z is a homotopy equivalence, we can find a homotopy inverse k for e_Z and a homotopy $H : Z \times I \rightarrow Z$ such that $H(-, 0) = \text{id}_Z$ and $H(-, 1) = k \circ e_Z$. Define a map

$$\rho : \text{holim}(X \xrightarrow{f} Z' \xleftarrow{g} Y) \rightarrow \text{holim}(X \xrightarrow{e_Z \circ f} Z \xleftarrow{e_Z \circ g} Y)$$

by the formula

$$(x, \gamma', y) \mapsto (x, H(f(x), -) * k(\gamma'(-)) * \overline{H(g(y), -)} : I \rightarrow Z, y)$$

where $*$ denotes concatenation of paths. The path concatenation is well defined because we have that $H(f(x), 1) = (k \circ e_Z \circ f)(x) = (k \circ \gamma')(0)$ and $k(\gamma'(1)) = k(e_Z(g(y))) = H(g(y), 1)$. This is well defined on the homotopy pullback because we have that

- $H(f(x), -) * k(\gamma'(-)) * \overline{H(g(y), -)}(0) = H(f(x), 0) = \text{id}_Z(f(x)) = f(x)$
- $H(f(x), -) * k(\gamma'(-)) * \overline{H(g(y), -)}(1) = H(g(y), 0) = \text{id}_Z(g(y)) = g(y)$

I claim that this map is the homotopy inverse to the map $\phi = \phi_{X,Z,Y}^{X',Z',Y'}$. We have that

$$\begin{aligned} \rho(\phi(x, \gamma, y)) &= \rho(x, e_Z \circ \gamma, y) \\ &= (x, H(f(x), -) * k(e_Z(\gamma(-)) * \overline{H(g(y), -)}, y) \end{aligned}$$

Now I claim that the middle path is homotopic to γ . For the first component of the concatenation, the path $H(f(x), t) : I \rightarrow Z$ can be contracted to $H(f(x), 0) = f(x) = \gamma(0)$ so you can homotope the traversal along $H(f(x), -)$ to the single point $f(x) = \gamma(0)$. For the third component of the concatenation, this is similar so we can homotope the traversal of $\overline{H(g(y), -)}$ to the single point $g(y) = \gamma(1)$. The middle part of the path is homotopic to γ because $k \circ e_Z$ is homotopic to id_Z . Thus we conclude. ■

We note that the above is still true if we replace homotopy equivalences by weak equivalences. Then the induced map between standard homotopy pushouts is a weak equivalence. This is given in [?].

7.2 Homotopy Pushout and Pullback Squares

Recall that we want our new pushout to be a homotopy invariant, not a homeomorphic invariant. Any homeomorphic spaces has one unique way of writing it down set theoretically up to bijection of the underlying set and the topology, but homotopy equivalent spaces are not determined by its set theoretic representation.

Definition 7.2.1 (Homotopy Commutative Square)

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that the following is (not necessarily commutative) diagram:

$$\begin{array}{ccc} W & \xrightarrow{f} & Y \\ \downarrow g & & \downarrow h \\ X & \xrightarrow{k} & Z \end{array}$$

We say that the square is homotopy commutative if there exists homotopy from $h \circ f$ to $k \circ g$.

Since homotopy is an equivalent relation on maps we see that the existence of a homotopy from $k \circ g$ to $h \circ f$ also gives us a homotopy commutative square.

Example 7.2.2

Consider the following square:

$$\begin{array}{ccc} \mathrm{holim}(X \rightarrow Z \leftarrow Y) & \xrightarrow{f} & Y \\ \downarrow g & & \downarrow h \\ X & \xrightarrow{k} & Z \end{array}$$

where the arrows from the homotopy pullback is given by projections. The diagram is not commutative because going through the two possible compositions only give points in Z that are connected by a path. However, it is homotopy commutative if we consider the homotopy $H : \mathrm{holim}(X \rightarrow Z \leftarrow Y) \times I \rightarrow Z$ defined by

$$H((x, \gamma, y), t) = \gamma(t)$$

The situation is dual with the standard homotopy pushout.

In particular, a commutative square is also homotopy commutative since we can take the constant homotopy between the two maps.

Definition 7.2.3 (Homotopy Pushout Square)

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that there is a homotopy commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g} & Y \\ \downarrow f & & \downarrow k \\ X & \xrightarrow{h} & Z \end{array}$$

whose homotopy from $h \circ f$ to $k \circ g$ is witnessed by $H : W \times I \rightarrow Z$. Define the map $\alpha : \mathrm{hocolim}(X \leftarrow W \rightarrow Y) \rightarrow Z$ by the formula

$$\alpha(u) = \begin{cases} h(u) & \text{if } u \in X \\ k(u) & \text{if } u \in Y \\ H(w, t) & \text{if } u = (w, t) \in W \times I \end{cases}$$

(Notice that it is well defined since $H(w, 0) = h(f(w))$ and $H(W, 1) = k(g(y))$).

- We say that the square is a homotopy pushout square if α is a weak equivalence.
- We say that the diagram is k -cocartesian if α is k -connected.

The idea is similar to standard pushouts in the sense that if we have a commutative square

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & Z \end{array}$$

then by the universal property of pushouts one would obtain a comparison map $W \rightarrow \lim(X \rightarrow Z \leftarrow Y)$. When this is a homeomorphism we call W a pushout of the diagram. Similarly we define homotopy pullback squares dually.

Definition 7.2.4 (Homotopy Pullback Square) Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that there is a homotopy commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g} & Y \\ \downarrow f & & \downarrow k \\ X & \xrightarrow{h} & Z \end{array}$$

whose homotopy from $h \circ f$ to $k \circ g$ is witnessed by $H : W \times I \rightarrow Z$. Let $\beta : W \rightarrow \text{holim}(X \rightarrow Z \leftarrow Y)$ be the map defined by

$$\beta(w) = (f(w), H(w, -), g(y))$$

(notice that the map is well defined since $H(w, 0) = h(f(w))$ and $H(w, 1) = k(g(y))$).

- We say that the diagram is a homotopy pullback if the β is a weak equivalence.
- We say that the diagram is k -cartesian if β is k -connected.

In the two definitions, I considered homotopy commutative square to be homotopy pushouts and homotopy pullbacks by considering the existence of a weak equivalence. One may ask to enforce the map to be a homotopy equivalence instead. While [?] only requires the map is a weak equivalence, [?] and [?] require them to be homotopy equivalences. Because every homotopy equivalence is a weak equivalence, using merely weak equivalences allows for a wider class of spaces to represent the same homotopy pushout or pullback. However the biggest drawback is that weak equivalences are not invertible in general, unless in specific examples such as CW complexes, in which case there is no difference between specifying the map to be a weak equivalence or a homotopy equivalence.

Suppose the following is a commutative square in $\mathbf{Top}$:

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & Z \end{array}$$

Notice that the map α factors through the limit because we are taking the homotopy recording this homotopy commutative square to be the constant homotopy. The same is true for the comparison map for homotopy pullbacks.

[?] only compares homotopy pushouts / pullbacks with commutative squares and so defines a square to homotopy pushout / pullback if the comparison map is a weak equivalence. However the notion is quite limiting since the simplest examples fail to be a homotopy pushout square. For instance, the square

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & \text{hocolim}(X \leftarrow W \rightarrow Y) \end{array}$$

is not even an honest commutative square.

Proposition 7.2.5

Let the following be a pullback square:

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & Z \end{array}$$

If one of $X \rightarrow Z$ or $Y \rightarrow Z$ is a fibration, then the square is a homotopy pullback square.

Proof By ??, if one of $X \rightarrow Z$ or $Y \rightarrow Z$ is a fibration, then the canonical map from the limit to the homotopy limit is a homotopy equivalence. Hence we obtain a weak equivalence $W \cong \lim(X \rightarrow Z \leftarrow Y) \rightarrow \operatorname{holim}(X \rightarrow Z \leftarrow Y)$. ■

The following proposition shows that recognizing homotopy pullback squares can be done through other factorizations of maps into weak equivalences and fibrations. The theorem can be found in section 13.3 of [?]. The following proof is original.

Proposition 7.2.6

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that there is a homotopy commutative diagram

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Z \end{array}$$

Then the square is a homotopy pullback square if and only if for any factorization of $Y \rightarrow Z$ into a weak equivalence $Y \rightarrow M$ and a fibration $M \rightarrow Z$, the induced map

$$W \rightarrow \lim(X \rightarrow Z \leftarrow M)$$

is a weak equivalence.

Proof Further factorize $Y \rightarrow M$ into a weak equivalence $Y \rightarrow E$ and fibration $E \rightarrow M$ so that we have a acyclic cofibration $Y \rightarrow E$ and a fibration $E \rightarrow Z$. This is possible by section 2 of [?], which in turn cites [?]. Then notice that we have the following commutative diagram:

$$\begin{array}{ccc} Y & \longrightarrow & P_g \\ \downarrow & & \downarrow \\ E & \longrightarrow & Z \end{array}$$

because they are both factorizations of $Y \rightarrow Z$, the diagram is commutative. Since $Y \rightarrow E$ is an acyclic cofibration and $P_g \rightarrow Z$ is a fibration, there exists a lift $E \rightarrow P_g$ making the above diagram commute by ??. Since $Y \rightarrow P_g$ and $Y \rightarrow E$ are weak equivalences, $E \rightarrow P_g$ is also a weak equivalence by the two out of three property in ??. Consider the following diagram:

$$\begin{array}{ccccc} W & \xrightarrow{\quad} & Y & & \\ & \searrow & \downarrow & & \\ & & \lim(P_f \rightarrow Z \leftarrow E) & \longrightarrow & E \\ & \searrow & \downarrow & & \downarrow \\ & & \lim(P_f \rightarrow Z \leftarrow P_g) & \longrightarrow & P_g \\ & \searrow & \downarrow & & \downarrow \\ X & \xrightarrow{\quad} & P_f & \longrightarrow & Z \end{array}$$

where the maps from W to the limits are induced by the universal property of pullbacks, and so the triangle with W and the two limits is a commutative diagram. By ??, $\lim(P_f \rightarrow Z \leftarrow E) \rightarrow \lim(P_f \rightarrow Z \leftarrow P_g)$ is a weak equivalence. By the two out of three property ??, $W \rightarrow \lim(P_f \rightarrow Z \leftarrow P_g) \cong \operatorname{holim}(X \rightarrow Z \leftarrow Y)$ is a weak equivalence if and only if $W \rightarrow \lim(P_f \rightarrow Z \leftarrow E)$ is a weak equivalence.

Finally, notice that since $Y \rightarrow E \rightarrow M$ is a factorization of the map $Y \rightarrow M$, we know that $E \rightarrow M$ is a weak equivalence by the two out of three property ??. By ?? we know that $\lim(P_f \rightarrow Z \leftarrow E) \rightarrow \lim(P_f \rightarrow Z \leftarrow M)$ is a weak equivalence. Therefore by the two out of three property ??, $W \rightarrow \lim(P_f \rightarrow Z \leftarrow E)$ is a weak equivalence if and only if $W \rightarrow \lim(P_f \rightarrow Z \leftarrow M)$ is a weak equivalence. This shows that $W \rightarrow \operatorname{holim}(X \rightarrow Z \leftarrow Y)$ is a weak equivalence if and only if $W \rightarrow \lim(P_f \rightarrow Z \leftarrow M)$ is a weak equivalence, proving our claim. ■

The following proposition is a generalization of the one given in [?] since it deals with homotopy commutative squares instead of strictly commutative squares. The proof is inspired by that in section 13.3 of [?]

Proposition 7.2.7

Consider the following homotopy commutative square:

$$\begin{array}{ccccc} X_1 & \longrightarrow & X_2 & \longrightarrow & X_3 \\ \downarrow & & \downarrow & & \downarrow \\ Y_1 & \longrightarrow & Y_2 & \longrightarrow & Y_3 \end{array}$$

in **Top**. Let the right square be a homotopy pullback square. Then the left square is a homotopy pullback if and only if the rectangle is a homotopy pullback square.

Proof Factorize $X_3 \rightarrow Y_3$ into a weak equivalence $X_3 \rightarrow P$ and a fibration $P \rightarrow Y_3$ where P is the mapping path space of the map $X_3 \rightarrow Y_3$. We then obtain the following commutative diagram:

$$\begin{array}{ccccc} X_1 & & X_2 & & X_3 \\ \downarrow & & \downarrow & & \downarrow \simeq \\ \lim(Y_1 \rightarrow Y_3 \leftarrow P) & \longrightarrow & \lim(Y_2 \rightarrow Y_3 \leftarrow P) & \longrightarrow & P \\ \downarrow & & \downarrow & & \downarrow \text{fib} \\ Y_1 & \longrightarrow & Y_2 & \longrightarrow & Y_3 \end{array}$$

where the two vertical maps on the top left is induced by the canonical map of the homotopy pullback square. We are given that $X_2 \rightarrow \lim(Y_2 \rightarrow Y_3 \leftarrow P)$ is a weak equivalence. By the pasting law of pullback squares ??, we know that the bottom left square is a pullback square. Since $P \rightarrow Y_3$ is a fibration and the bottom right square is a pullback square, the middle lower vertical map is a fibration by ??.

Notice that the map $X_2 \rightarrow Y_2$ is factored into a weak equivalence $X_2 \rightarrow \lim(Y_2 \rightarrow Y_3 \leftarrow P)$ followed by a fibration $\lim(Y_2 \rightarrow Y_3 \leftarrow P) \rightarrow Y_2$. Moreover by the pasting law, the bottom left square is a pullback. Therefore the left square in the original diagram is a homotopy pullback square if and only if $X_1 \rightarrow \lim(Y_1 \rightarrow Y_3 \leftarrow P)$ is a weak equivalence by ??. But also the same map is a weak equivalence if and only if the outer rectangle is a homotopy pullback square by definition. Hence we conclude. ■

The dual version for homotopy pushouts is also true. Namely, if the following is homotopy commutative:

$$\begin{array}{ccccc} X_1 & \longrightarrow & X_2 & \longrightarrow & X_3 \\ \downarrow & & \downarrow & & \downarrow \\ Y_1 & \longrightarrow & Y_2 & \longrightarrow & Y_3 \end{array}$$

Assume that the left square is a homotopy pushout square, then the right square is a homotopy pushout square if and only if the left square is a homotopy pushout square. A proof in the case where the diagram is strictly commutative can be found in proposition 3.7.26 of [?]. In the proof of ?? we will use the strictly commutative version.

Proposition 7.2.8

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that there is a homotopy commutative diagram

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Z \end{array}$$

Then the following are true.

- If the square is a homotopy pullback, and $Y \rightarrow Z$ is a weak equivalence, then $W \rightarrow X$ is a

weak equivalence.

- If $Y \rightarrow Z$ and $W \rightarrow X$ are weak equivalence, then the square is a homotopy pullback.

Proof The important square to consider is the following pullback square:

$$\begin{array}{ccc} \mathrm{holim}(X \rightarrow Z \leftarrow Y) & \longrightarrow & Y \\ \downarrow & & \downarrow \\ P_f & \longrightarrow & Z \end{array}$$

Since $Y \rightarrow Z$ is a weak equivalence and $P_f \rightarrow Z$ is a fibration, ?? implies that $\mathrm{holim}(X \rightarrow Z \leftarrow Y) \rightarrow P_f$ is a weak equivalence. Since this map is given by the composition $\mathrm{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X \rightarrow P_f$ and $X \rightarrow P_f$ is a weak equivalence, the two out of three property ?? gives that $\mathrm{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X$ is a weak equivalence. Since we have a weak equivalence $W \rightarrow \mathrm{holim}(X \rightarrow Z \leftarrow Y)$, we again apply the two out of three property ?? so that the composition $W \rightarrow X$ is a weak equivalence.

Conversely, if $Y \rightarrow Z$ and $W \rightarrow X$ are weak equivalence, then the same reasoning shows that $\mathrm{holim}(X \rightarrow Z \leftarrow Y) \rightarrow P_f$ is a weak equivalence and hence $\mathrm{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X$ is a weak equivalence by two out of three ?. Since $W \rightarrow X$ is also a weak equivalence, by two out of three ?? we conclude that $W \rightarrow \mathrm{holim}(X \rightarrow Z \leftarrow Y)$ is a weak equivalence and so the square in question is a homotopy pullback. ■

The statement dualizes into a version involving homotopy pushout squares: Namely if one has a square:

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Z \end{array}$$

and $W \rightarrow X$ is a weak equivalence, then the square is a homotopy pushout if and only if $Y \rightarrow Z$ is a weak equivalence. The proof of this can be found in [?].

Proposition 7.2.9

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that following is a homotopy commutative square

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow h & & \downarrow g \\ X & \xrightarrow{f} & Z \end{array}$$

The square is a homotopy pullback if and only if for all $x \in X$, the map

$$\mathrm{hofib}_x(h) \rightarrow \mathrm{hofib}_{f(x)}(g)$$

is a weak equivalence.

Proof Consider the following homotopy commutative diagram.

$$\begin{array}{ccccccc} \mathrm{hofib}_{\gamma(f(x))}(\alpha) & \longrightarrow & \mathrm{hofib}_x(h) & \longrightarrow & W & & \\ \downarrow & & \downarrow \beta & & \downarrow \alpha & & \\ * & \longrightarrow & \mathrm{hofib}_{f(x)}(g) & \xrightarrow{\gamma} & \mathrm{holim}(X \rightarrow Z \leftarrow Y) & \longrightarrow & Y \\ & & \downarrow & & \downarrow & & \downarrow g \\ & & * & \longrightarrow & X & \longrightarrow & Z \end{array}$$

Notice that by definition, the map $W \rightarrow \mathrm{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X$ is precisely the map h . By repeated application of ?? we conclude that all four squares above are homotopy pullback squares.

If α is a weak equivalence, then since the top wide rectangle is a homotopy pullback we have $\text{hofib}_{\gamma(f(x))}(\alpha)$ is weakly equivalent to $*$. But the top left square is a homotopy pullback hence $\text{holim}_x(h)$ is weakly equivalent to $\text{holim}_{f(x)}(g)$. Conversely, suppose that β is a weak equivalence for all x . Since the top left square is a homotopy pullback, this implies that $\text{hofib}_{\gamma(f(x))}(\alpha)$ is weakly contractible for all x . In particular it is n -connected for all n . Then this implies that α is $(n+1)$ -connected for all n . Hence α is a weak equivalence. ■

The above proposition is also true if we replace weak equivalence with k -cartesian and can be found in [?].

Proposition 7.2.10

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that there is a homotopy pushout square

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & Z \end{array}$$

Then there exists a pushout square

$$\begin{array}{ccc} W' & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ X' & \longrightarrow & Z' \end{array}$$

such that $W' \rightarrow X'$, $W' \rightarrow Y'$, $\text{colim}(X' \leftarrow W' \rightarrow Y') \rightarrow Z$ are cofibrations, and a map of squares

$$\begin{array}{ccccc} W' & \longrightarrow & Y' & & \\ \downarrow & \searrow & \downarrow & \searrow & \\ & W & \longrightarrow & Y & \\ \downarrow & & \downarrow & & \downarrow \\ X' & \longrightarrow & Z' & & \\ \downarrow & \searrow & \downarrow & \searrow & \\ & X & \longrightarrow & Z & \end{array}$$

where the maps $W' \rightarrow W$, $X' \rightarrow X$, $Y' \rightarrow Y$, $Z' \rightarrow Z$ are homotopy equivalences.

Proof Take $W' = W$, Y' the mapping cylinder of $W \rightarrow Y$ and X' the mapping cylinder of $W \rightarrow X$. Take Z' to be the mapping cylinder of

$$\text{colim}(M_{W \rightarrow X} \leftarrow W \rightarrow M_{W \rightarrow Y}) \cong \text{holim}(X \leftarrow W \rightarrow Y) \rightarrow Z$$

It is clear that $W' \rightarrow X'$ and $W' \rightarrow Y'$ are cofibrations. Moreover, $\text{colim}(X' \leftarrow W' \rightarrow Y') \rightarrow Z'$ is also a cofibration is a same reasoning. ■

Proposition 7.2.11

Let $W, X, Y, Z \in \mathbf{Top}$ be spaces such that there is a commutative homotopy pullback square

$$\begin{array}{ccc} W & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X & \longrightarrow & Z \end{array}$$

Then the homotopy fiber of the induced map $W \rightarrow X \times Y$ is weakly equivalent to ΩZ . Moreover, the fiber sequence as in ?? induces a long exact sequence of homotopy groups:

$$\cdots \longrightarrow \pi_{n+1}(Z) \longrightarrow \pi_n(W) \longrightarrow \pi_n(X) \times \pi_n(Y) \longrightarrow \cdots \longrightarrow \pi_0(X) \times \pi_0(Y)$$

Proof Consider the following commutative diagram:

$$\begin{array}{ccccc}
 X & \longrightarrow & Z & \longleftarrow & Y \\
 \text{id} \downarrow & & \downarrow & & \downarrow \text{id} \\
 X & \longrightarrow & * & \longleftarrow & Y \\
 \uparrow & & \uparrow & & \uparrow \\
 * & \longrightarrow & * & \longleftarrow & *
 \end{array}$$

Using ?? and ??, we see that the homotopy limit of the homotopy limit of the rows is precisely the homotopy fiber of the map $\text{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X \times Y$. The homotopy limit of the homotopy limit of the columns is homotopy equivalent to $\text{holim}(* \rightarrow Z \leftarrow *) \cong \Omega Z$ using ?? and ??. Now the maps $W \rightarrow \text{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X, Y$ is precisely the maps $W \rightarrow X, Y$ in the original diagram by definition. Therefore we obtain the following commutative diagram:

$$\begin{array}{ccc}
 W & \longrightarrow & \text{holim}(X \rightarrow Z \leftarrow Y) \\
 \downarrow & & \downarrow \\
 X \times Y & \xrightarrow{\text{id}} & X \times Y
 \end{array}$$

Since the top horizontal arrow is a weak equivalence, we conclude that this square is a homotopy pullback square by ??. Then there is a weak equivalence

$$\text{hofib}_{(x,y)}(W \rightarrow X \times Y) \rightarrow \text{hofib}_{(x,y)}(\text{holim}(X \rightarrow Z \leftarrow Y) \rightarrow X \times Y)$$

by ??. Since the latter space is homotopy equivalent to ΩZ , the long exact sequence in ?? completes the proof. ■

7.3 The Blakers-Massey Theorem for Squares

The Blakers-Massey theorem is a direct generalization of the homotopy excision theorem. Its proof takes a similar form to the homotopy excision theorem. Let us recall some definitions used.

Definition 7.3.1 ((Degenerative) Cubes) Let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$. Let $\delta > 0$. Let $L \subseteq \{1, \dots, n\}$. A cube in $\mathbb{R}^n$ is a set of the form

$$W = W(a, \delta, L) = \{x \in \mathbb{R}^n \mid a_i \leq x \leq a_i + \delta \text{ for } i \in L \text{ and } x_i = a_i \text{ for } i \notin L\}$$

The notation is making the object more complicated than what it should look like. $a \in \mathbb{R}^n$ refers to the bottom left coordinate of the cube. δ is the length of the cube and L refers to the number of non-degenerate faces of the cube. In particular, any cube in $\mathbb{R}^n$ is homeomorphic to the standard cube I^k for some $k \leq n$.

- When $n = 3$, $W(0, 1, \{1, 2\})$ is the unit square on the xy -plane.
- When $n = 3$, $W(0, 1, \{1, 2, 3\})$ is the unit cube.
- When $n = 4$, $W(0, 1, \{1, 2, 3\})$ is the unit cube with nonzero first three coordinates and zero otherwise.

Definition 7.3.2 (Special Sub-cube of a Cube) Let $W = W(a, \delta, L)$ be a cube in $\mathbb{R}^n$. Let $j = 1$ or 2 . Suppose that $1 \leq p \leq |L|$. Define

$$K_p^j(W) = \left\{ (x_1, \dots, x_n) \in W \mid \frac{\delta(j-1)}{2} + a_i < x_i < \frac{\delta j}{2} + a_i \text{ for at least } p \text{ values of } i \in L \right\}$$

Again the notation is making the object more complicated. Taking $W = W(0, 1, \{1, 2, 3\}) = I^3$ in $\mathbb{R}^3$, we have

- $K_3^1(W) = W(0, 1/2, \{1, 2, 3\})$ is one eighth of the cube I^3 with bottom left corner at the origin.

- $K_3^2(W) = W((1/2, 1/2, 1/2), 1/2, \{1, 2, 3\})$ is one eighth of the cube I^3 with bottom left corner at $(1/2, 1/2, 1/2)$
- $K_2^1(W)$ allows for one coordinate to go beyond the bottom left one eighth of the cube, and is the union four of the $1/8$ -sub-cubes that are adjacent to the xy -plane, the yz -plane and the xz -plane.
- $K_1^1(W)$ allows for two coordinate to go beyond the bottom left one eighth of the cube, and is equal to $W \setminus K_3^2(W)$.
- $K_0^1(W)$ allows for all coordinate to go beyond the bottom left one eighth of the cube, so the condition becomes vacuous and is equal to W .

Summarizing, we think of $K_p^j(W)$ as follows. Subdivide the $|L|$ -dimensional cube into $2^{|L|}$ sub-cubes of equal volume. K_p^1 is the union of a number of sub-cubes closest to the bottom left sub-cube. K_p^2 is the union of a number of sub-cubes closed to the upper right sub-cube.

Lemma 7.3.3 Let Y be a space. Let $B \subseteq Y$ be a subspace of Y . Let $W = W(a, \partial, L)$ be a cube in $\mathbb{R}^n$. Let $f : W \rightarrow Y$ be a map. Let $j = 1$ or 2 . Suppose that there exists some $p \leq |L|$ such that

$$f^{-1}(B) \cap C \subset K_p^j(C)$$

for all cubes $C \subset \partial W$. Then there exists a map $g : W \rightarrow Y$ such that $g \stackrel{\partial W}{\simeq} f$ and

$$g^{-1}(B) \subset K_p^j(W)$$

Proof (Proof by Munson in Cubical Homotopy Theory) Firstly, notice that any cube W is homeomorphic to I^n for some n , so we can just prove the statement for when $W = I^n$. In this case, our parameters of the cube is given by $I^n = W(a = 0, \delta = 1, L = \{1, \dots, n\})$ and our $K_p^j(W)$ is given by

$$K_p^j(W) = \left\{ (x_1, \dots, x_n) \in C \mid \frac{j-1}{2} < x_i < \frac{j}{2} \text{ for at least } p \text{ values of } i \in \{1, \dots, n\} \right\}$$

Let p_j be the center of the sub-cube $[\frac{j-1}{2}, \frac{j}{2}]^n$ inside I^n for $j = 1, 2$. Let R be a ray with starting point p_j . Let $P(R, p_j)$ be the intersection of R and $\partial [\frac{j-1}{2}, \frac{j}{2}]^n$. Let $Q(R, p_j)$ be the intersection of R and ∂I^n . By construction, the points p_j , $P(R, p_j)$ and $Q(R, p_j)$ are collinear with $P(R, p_j)$ always being the mid point and $P(R, p_j)$ is possibly equal to $Q(R, p_j)$. Being a line, we can define a linear homotopy from the line $[p_j, P(R, p_j)]$ to the line $[p_j, Q(R, p_j)]$ that fixes the point p_j and sends $P(R, p_j)$ to $Q(R, p_j)$. Denote the homotopy by $h(y, t)$ for $y \in [p_j, P(R, p_j)]$ and t the time variable.

Now we can define a homotopy $H_j : I^n \times I \rightarrow I^n$ as follows: For each $y \in I^n$, there exists a unique ray R starting at p_j and passing through y . Then we obtain a homotopy h from $[p_j, P(R, p_j)]$ to $[p_j, Q(R, p_j)]$ as above. Define $H(y, t) = h(y, t)$. It is clear that $H_j(q, 1) = q$ for all $q \in \partial I^n$ so that H_j is a homotopy from the identity, relative to the boundary ∂I^n .

Let $g = f \circ H_j(-, 1)$. From the properties of the homotopy H_j , we notice that $f \circ H_j : I^n \times I \rightarrow Y$ is a homotopy from $f \circ H(-, 0) = f \circ \text{id} = f$ to $f \circ H(-, 1) = g$, relative to the boundary ∂I^n . Thus we now have a homotopy from f to a map g relative to the boundary. It remains to show that $g^{-1}(B) \subset K_p^j(C)$.

Let $z = (z_1, \dots, z_n) \in g^{-1}(B)$. If $z \in [\frac{j-1}{2}, \frac{j}{2}]^n$ then clearly $z \in K_p^j(C)$ is true. So suppose instead that $z = (z_1, \dots, z_n) \in g^{-1}(B)$ satisfies the fact that either $z_a \geq \frac{j}{2}$ or $z_b \leq \frac{j-1}{2}$ for $1 \leq a, b \leq n$. Let R be the ray from p_j passing through z . Then the condition on z means that $z \in$

$[P(R, p_j), Q(R, p_j)]$. Hence under the homotopy H , z is mapped to ∂I^n . But ∂I^n is a union $n - 1$ dimensional faces of I^n which are cubes. So $H(z, 1)$ lies in some cube $C \subseteq \partial I^n$. By construction of g , $g(z) = f(H(z, 1))$ and $g(z) \in B$ implies that $H(z, 1) \in f^{-1}(B)$. Then $H(z, 1) \in f^{-1}(B)$ and $H(z, 1) \in C$ implies that

$$H(z, 1) \in f^{-1}(B) \cap C \subseteq K_p^j(C)$$

by the assumption on f . Write $H(z, 1) = (w_1, \dots, w_n) \in \partial I^n$. This means that $\frac{j-1}{2} < w_i < \frac{j}{2}$ for at least p of the coordinates of $H(z, 1)$.

Now the ray starting at p_j and passing through z is parametrized by the line $p_j + t(z - p_j)$ for $t \geq 0$. Since $H(z, 1)$ lies behind the two points z and p_j , we can write $H(z, 1) = p_j - t_0(z - p_j)$ for some $t_0 \geq 1$. By definition, p_j is the point given in coordinates by $(\frac{2j-1}{4}, \dots, \frac{2j-1}{4})$. Hence the i th coordinate of $H(z, 1)$ can be written as

$$w_i = \frac{2j-1}{4} + t_0 \left(z_i - \frac{2j-1}{4} \right)$$

Recall that in the previous paragraph we found that $\frac{j-1}{2} < w_i < \frac{j}{2}$ for at least p of the coordinates of $H(z, 1)$. Substituting w_i into the inequality and simplifying gives

$$-\frac{1}{4t_0} + \frac{2j-1}{4} < z_i < \frac{1}{4t_0} + \frac{2j-1}{4}$$

Since $t_0 \geq 1$, we get

$$-\frac{1}{4} + \frac{2j-1}{4} < -\frac{1}{4t_0} + \frac{2j-1}{4} < z_i < \frac{1}{4t_0} + \frac{2j-1}{4} < \frac{1}{4} + \frac{2j-1}{4}$$

The leftmost and rightmost terms bound z_i between $\frac{j-1}{2}$ and $\frac{j}{2}$ for at least p amount of coordinates z_i of z . Hence $z \in K_p^j(C)$. This completes the proof. $\blacksquare$

Lemma 7.3.4 Let X be a space. Let $X_0, X_1, X_2 \subseteq X$ be subspaces of X such that

$$X = X_1 \cup X_2$$

and $X_0 = X_1 \cap X_2$ is non-empty. Assume that for each $i = 1, 2$, (X_i, X_0) is k_i -connected with $k_i \geq 0$. Let $f : I^n \rightarrow X$ be a map. Let

$$I^n = \bigcup_k W_k$$

be the decomposition of I^n into cubes W_k such that $f(W_k) \subseteq X_i$ for one of $i = 0, 1, 2$ by the Lebesgue covering lemma. Then there exists a homotopy

$$H : I^n \times I \rightarrow X$$

such that the following are true.

- $f(-) = H(-, 0)$
- If $f(W) \subset X_i$, then $H(W, t) \subset X_i$ for all $t \in I$.
- If $f(W) \subset X_0$, then $H(W, t) = f(W)$ for all $t \in I$.
- If $f(W) \subset X_i$, then $((H(-, 1))^{-1}(X_i \setminus X_0)) \cap W \subset K_{k_i+1}^i(W)$.

Proof Let C^d be the union of all cubes of dimension $\leq d$. We induct on d , the existence of such a homotopy $H : C^d \times I \rightarrow X$ that holds the required conditions true for all cubes W with dimension $\leq d$.

We first construct the homotopy for all cubes of dimension 0. When $\dim(W) = 0$, there are two cases:

- If $f(W) \subset X_0$, define $H|_{W \times I}$ by $H(w, t) = f(w)$
- If $f(W) \subset X_j$ and $f(W) \not\subset X_i$ for $1 \leq i \neq j \leq 2$, (X_j, X_0) is $(k_j \geq 0)$ -connected implies that there exists a path $\gamma : I \rightarrow X$ from $f(W)$ to a point in X_0 . Define $H|_{W \times I}$ by $H(w, t) = \gamma(t)$ (again $W = \{w\}$ is a one point set).

Thus we now have a well defined map $H : C^0 \times I \rightarrow X$. We need to show that this map satisfies the required conditions.

- For each $z \in C^0$, either $H(z, 0) = f(z)$ from the first case or $H(z, 0) = \gamma(0) = f(z)$.
- If $f(W) \subset X_i$, then by construction $H(W, t) \subset X_i$ from the second case.
- If $f(W) \subset X_0$, then $H(W, t) = f(W)$ by the first case.
- $K_{k_i+1}^i(W) = \{w\}$ is a one point set and $(H(-, 1))^{-1}(X_i \setminus X_0) \cap W \subseteq W$ means that this condition is satisfied.

Now H is built on three pieces: the union of cubes landing in X_i for $i = 0, 1, 2$. The second and third conditions guarantee that each of the three pieces define a homotopy on each piece respectively. Since $\partial W \hookrightarrow W$ is a cofibration, we can extend these pieces of homotopy from 0-dimensional cubes to 1-dimensional. Recursively we are able to define a homotopy for all cubes of all dimensions inside I^n that satisfy the first three conditions.

Therefore now we can invoke the inductive hypothesis, so that there exists a homotopy from f so that the new function satisfy all our required conditions for all cubes of dimension $< d$. With abuse of notation, call the restriction of our newly acquired function to C^{d-1} also by the name f . Let W be a cube of dimension d .

- If $f(W) \subseteq X_0$, define $H|_{W \times I}$ by $H(w, t) = f(w)$
- If $f(W) \subset X_1$ and $f(W) \not\subset X_2$ and $\dim(W) = d \leq k_1$, (X_j, X_0) is $(k_j \geq 0)$ -connected implies there exists a homotopy $K : W \times I \rightarrow X$ from f relative to ∂W such that $K(W, 1) \subseteq X_0$. Define $H|_{W \times I}$ by $H = K$.
- If $f(W) \subset X_1$ and $f(W) \not\subset X_2$ and $\dim(W) = d > k_1$, then by induction we have

$$f^{-1}(X_1 \setminus X_0) \cap W' \subset K_d^1(W') \subset K_{k_1+1}^1(W')$$

for all $W' \subset \partial W$ (induction is applicable since $\dim(W') < \dim(W)$). By the above lemma, there exists a map $g : W \rightarrow X$ such that g and f are homotopic relative to ∂W such that $g^{-1}(X_1 \setminus X_0) \subset K_{k_1+1}^1(W)$. Call this homotopy from f to g by $R : W \times I \rightarrow X$. Then we define $H|_{W \times I}$ by $H = R$.

(WLOG the cases where X_1 is swapped with X_2 and k_1 is swapped with k_2 and $K_p^1(W)$ is swapped with $K_p^2(W)$ in the last two sub-cases has a symmetrical argument). Finally we show that our required conditions are satisfied.

- In all cases, $H(-, 0) = f$ as one can immediately see.
- The second condition holds for all cubes of dimension $< d$ by inductive hypothesis. It also holds for our first and second case since $X_0 \subset X_1, X_2$. For the third case, H is a homotopy relative to ∂W . Since by induction hypothesis $f(\partial W) \subseteq X_1$, we also have $g(\partial W) \subseteq X_1$. Since g is continuous then $H(W, 1) = g(W) \subseteq X_1$.
- The third condition holds for all cubes of dimension $< d$ by inductive hypothesis, and holds true for all cubes of dimension d by the first case.
- The fourth condition holds true by our argument in the third case, and is vacuously true in the second case since $H(W, 1) \subseteq X_0$ implies that $(H(-, 1))^{-1}(X_1 \setminus X_0) = \emptyset$.

Thus the proof is complete. ■

We can now prove a weaker version of Blakers-Massey theorem.

Proposition 7.3.5 Let X be a space. Let e^{d_i} be a cell of dimension d_i for $i = 1, 2$. Then the following diagram

$$\begin{array}{ccc} X & \longrightarrow & X \cup e^{d_1} \\ \downarrow & & \downarrow \\ X \cup e^{d_2} & \longrightarrow & X \cup e^{d_1} \cup e^{d_2} \end{array}$$

given by inclusion maps is $(d_1 + d_2 - 3)$ -cartesian.

Proof (Proof is by Munson in Cubical Homotopy Theory) Let $p_1 \in e^{d_1}$ and $p_2 \in e^{d_2}$ be interior points. Since $X \cup e^{d_2}$ is weakly equivalent to $X \cup e^{d_1} \cup e^{d_2} \setminus \{p_1\}$ by inclusion (and similarly for $X \cup e^{d_1}$), the above square admits a weak equivalence to the following square:

$$\begin{array}{ccc} X \cup e^{d_1} \cup e^{d_2} \setminus \{p_1, p_2\} & \hookrightarrow & X \cup e^{d_1} \cup e^{d_2} \setminus \{p_2\} \\ \downarrow & & \downarrow \\ X \cup e^{d_1} \cup e^{d_2} \setminus \{p_1\} & \hookrightarrow & X \cup e^{d_1} \cup e^{d_2} \end{array}$$

Let $Y = X \cup e^{d_1} \cup e^{d_2}$. By thm???? to show that the square is $(d_1 + d_2 - 3)$ -cartesian is the same as showing the map

$$\mathrm{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) \rightarrow \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y)$$

is $(d_1 + d_2 - 3)$ -cartesian. Let

$$C = \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y \setminus \{p_2\}) \simeq *$$

Now the intersections below can be calculated to give

$$C \cap \mathrm{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) = \mathrm{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1, p_2\})$$

and

$$C \cap \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y) = \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y \setminus \{p_2\}) \simeq *$$

(because C and the homotopy fibers are all subspaces of $Y \times \mathrm{Map}(I, Y)$). By thm??? we conclude that the square

$$\begin{array}{ccc} C \cap \mathrm{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) & \hookrightarrow & \mathrm{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) \\ \downarrow & & \downarrow \\ C & \hookrightarrow & C \cup \mathrm{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) \end{array}$$

is a homotopy pullback. Since the left two terms are contractible, it is a weak equivalence. By thm??? we conclude that the map on the right is a weak equivalence. Similarly, we can consider the diagram

$$\begin{array}{ccc} C \cap \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y) & \hookrightarrow & \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y) \\ \downarrow & & \downarrow \\ C & \hookrightarrow & C \cup \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y) \end{array}$$

which is a homotopy pullback. And the fact that the terms on the left are contractible imply that the map on the right (which is in fact equal)

$$\mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y) \xrightarrow{=} \mathrm{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y)$$

is a weak equivalence by the same theorem. We now have a chain of maps

$$\begin{array}{ccc}
\text{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) & \xleftarrow{\text{weak eq.}} & C \cup \text{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}) \\
& \nwarrow & \\
C \cup \text{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y) & \xrightarrow{\text{weak eq.}} & \text{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y)
\end{array}$$

where the middle map is inclusion. It remains to show that the middle map is $(d_1 + d_2 - 3)$ -connected. This is the same as showing that the pair

$$(C \cup \text{hofiber}_y(Y \setminus \{p_2\} \rightarrow Y), C \cup \text{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\}))$$

is $(d_1 + d_2 - 3)$ -connected.

To simplify notations let us write the pair as (A, B) . Let $\phi : (I^n, \partial I^n) \rightarrow (A, B)$ be a map. Recall that $A = \{(x, \phi) \in Y \times \text{Map}(I, Y) \mid \phi(0) = y, \phi(1) = x\}$. The first variable is determined by the end point of ϕ so giving a map $I^n \rightarrow A$ is the same as giving a map $I^n \rightarrow \text{Map}(I, Y)$ for which all paths in the image has starting point y and ending point in $Y \setminus \{p_2\}$. By the hom-product adjunction, this is equivalent to giving a map $\psi : I^n \times I \rightarrow Y$ such that $\psi(z, 0) = y$ is the base point and $\psi(z, 1) \in Y \setminus \{p_2\}$. Similarly, we can consider the map $\phi : \partial I^n \rightarrow B$ and deduce that $\phi(z)$ is a path lying entirely in the codomain of the map of the homotopy fiber $C = \text{hofib}_y(Y \setminus \{p_2\} \rightarrow Y \setminus \{p_2\})$ or it is a path lying entirely in the codomain of the map of the homotopy fiber $\text{hofiber}_y(Y \setminus \{p_1, p_2\} \rightarrow Y \setminus \{p_1\})$. By the same adjunction we conclude that our ψ above must also satisfy for any fixed $z \in \partial I^n$, $\psi(z, t)$ lies entirely in either $Y \setminus \{p_1\}$ or $Y \setminus \{p_2\}$ (and conversely these information give a map $(I^n, \partial I^n) \rightarrow (A, B)$ by the adjunction).

To summarize: we have a map

$$\psi : I^n \times I \rightarrow Y$$

such that

- $\psi(z, 0) = y$ is the base point for all $z \in I^n$.
- $\psi(z, 1) \in Y \setminus \{p_2\}$ for all $z \in I^n$.
- For any fixed $z \in \partial I^n$, $\psi(z, t)$ lies entirely in $Y \setminus \{p_1\}$ or $Y \setminus \{p_2\}$ for varying t .

The goal is to make a homotopy from ψ to a map whose third condition holds for any $z \in I^n$ when $n \leq d_1 + d_2 - 3$. Then passing through the adjunction again we see that our original map $(I^n, \partial I^n) \rightarrow (A, B)$ is homotopic to the constant map as required. Apply 5.1.4 to obtain a homotopy $H : I^n \times I \times I \rightarrow Y$ from ψ to a new map $\eta : I^n \times I \rightarrow Y$, such that we have a decomposition of $I^n \times I$ into cubes W and the following are true.

1. $\psi(W) \subset Y \setminus \{p_2\}$ implies $H(W, r) \subset Y \setminus \{p_2\}$ for all $r \in I$.
2. $\psi(W) \subset Y \setminus \{p_1\}$ implies $H(W, r) \subset Y \setminus \{p_1\}$ for all $r \in I$.
3. $\psi(W) \subset Y \setminus \{p_1, p_2\}$ implies $H(W, r) = \psi(W)$ for all $r \in I$.
4. $\psi(W) \subset Y \setminus \{p_2\}$ then $(H(-, 1)^{-1}(\{p_1\})) \cap W \subset K_{d_1}^1(W)$
5. $\psi(W) \subset Y \setminus \{p_1\}$ then $(H(-, 1)^{-1}(\{p_2\})) \cap W \subset K_{d_2}^2(W)$

We claim that $H(z, t, r)$ satisfies the three bullet points for all fixed r .

Firstly, we already know that $\psi(z, 0) = y$ is the base point for all $z \in I^n$. So for all cubes $W \subseteq I^n \times \{0\}$, we have $\psi(W) = \{y\} \subset Y \setminus \{p_1, p_2\}$. By 3., we conclude that $H(W, r) = \psi(W) = \{y\}$. Secondly, we know that $\psi(z, 1) \in Y \setminus \{p_2\}$ for all $z \in I^n$. So for all cubes $W \subseteq I^n \times \{1\}$, we have $\psi(W) \subset Y \setminus \{p_2\}$. By 1., we conclude that $H(W, r) \subset Y \setminus \{p_2\}$ for all $r \in I$. Finally, according to the third bullet point, $\psi(z, t)$ lies entirely in $Y \setminus \{p_1\}$ or $Y \setminus \{p_2\}$ for $z \in \partial I^n$ WLOG lets say it lies entirely in $Y \setminus \{p_i\}$. Choose cubes $W_1, \dots, W_k$ in the decomposition of $I^n \times I$ so that it forms a minimal cover for $\{z\} \times I \subset W_1 \cup \dots \cup W_k$. By definition of the decomposition, these

cubes firstly contain at least one point in $\{z\} \times I$, and $\psi(\{z\} \times I) \subset Y \setminus \{p_j\}$ implies that $\psi(W_1), \dots, \psi(W_k) \subset Y \setminus \{p_j\}$. By 1., we conclude that $H(W_1, r), \dots, H(W_k, r) \subset Y \setminus \{p_j\}$ so that $H(W_1 \cup \dots \cup W_k, r) \subset Y \setminus \{p_j\}$.

It remains to show that $\eta(-, -) = H(-, -, 1)$ satisfies the stronger condition of the third bullet point as desired. Let $n \leq d_1 + d_2 - 3$. We want to show that $\eta(z, I) \subset Y \setminus \{p_j\}$ for some j . I claim that this is equivalent to saying

$$\text{proj}(\eta^{-1}(\{p_1\})) \cap \text{proj}(\eta^{-1}(\{p_2\})) = \emptyset$$

where proj is the projection to the first coordinate. Indeed if $\eta(z, I)$ always lie inside one of $Y \setminus \{p_j\}$, $j = 1, 2$, then $\text{proj}(\eta^{-1}(\{p_1\})) = \{z \in I^n \mid \eta(z, I) \subset Y \setminus \{p_1\}\}$ and similarly for the other projection so that their intersection is empty. Conversely if one of $\eta(z, I)$ does not entirely lie in $Y \setminus \{p_2\}$ then the intersection is non-empty.

So it suffices to prove that the intersection given above is empty. So suppose it is non-empty with an element z_0 . Then there exists $t_1, t_2 \in I$ such that $\eta(z_0, t_1) \in Y \setminus \{p_2\}$ and $\eta(z_0, t_2) \in Y \setminus \{p_1\}$. Choose cubes $W_1 = W(a_1, \delta_1, L_1), W_2 = W(a_2, \delta_2, L_2)$ in the given decomposition of $I^n \times I$ so that $(z_0, t_1) \in W_1$ and $(z_0, t_2) \in W_2$. By 4. and 5. we have $(z_0, t_1) \in \eta^{-1}(\{p_1\}) \cap W_1 \subset K_{d_1}^1(W_1)$ and similarly for (z_0, t_2) . Then (z_0, t_j) has at least d_j coordinates satisfying the inequalities to lie in $K_{d_j}^1$. Hence $z_0 = \text{proj}(z_0, t)$ has at least $d_j - 1$ coordinates satisfying those inequalities for each $j = 1, 2$. For each j , $\text{proj}(W_j)$ is a cube containing z_0 . Subdivide the cubes W_1 and W_2 further so that $\text{proj}(W_1) = \text{proj}(W_2)$. Since $\text{proj}(z_0, t) = z_0$, this means that z_0 has at least $d_j - 1$ coordinates satisfying the inequalities of $K_{d_1}^1(W_1)$ and $K_{d_2}^2(W_2)$. But notice that the inequalities of $K_{d_1}^1(W_1)$ and $K_{d_2}^2(W_2)$ are disjoint (one concerns whether the points are at the front of the cube, the other at the back). So these conditions are disjoint and z_0 must have at least $d_1 + d_2 - 2$ conditions on its coordinates. This is impossible if z_0 has less than $d_1 + d_2 - 3$ coordinates. Hence we are done.

Notice that the proof required $d_1, d_2 \geq 1$. Assume WLOG that $d_2 = 0$, then right from the beginning we are considering the map of homotopy fibers

$$\text{hofiber}_y(X \rightarrow X \cup e^{d_2}) \rightarrow \text{hofiber}_y(X \cup e^{d_1} \rightarrow X \cup e^{d_1} \cup e^{d_2})$$

where $X \cup e^{d_1}$ is now the disjoint union of X with a base point. Then $\text{hofiber}_y(X \rightarrow X \amalg *)$ consists of pairs $(x, \phi) \in X \times \text{Map}(I, X \amalg *)$ such that $\phi(0) \in X$ and $\phi(1) = *$. But $*$ is disjoint from X means that no pairs satisfy this conditions and the homotopy fiber is the empty set. Similarly for the target homotopy fiber. Hence the map of homotopy fibers is the identity and it is trivially true. The proof is similar for $d_1 = 0$. ■

Theorem 7.3.6 (Blakers-Massey Theorem for Squares) Let $X_0, X_1, X_2, X_{12} \in \mathbf{CGWH}$ be spaces such that the square

$$\begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_{12} \end{array}$$

is a homotopy pushout. Suppose the map $X_0 \rightarrow X_i$ is k_i -connected for $i = 1, 2$. Then the diagram is $(k_1 + k_2 - 1)$ -cartesian. Explicitly, this means that

$$\alpha : X_0 \rightarrow \text{holim}(X_1 \rightarrow X_{12} \leftarrow X_2)$$

is $(k_1 + k_2 - 1)$ -connected.

Proof



This theorem directly generalizes the homotopy excision theorem in the following way. For X a CW complex and A, B two subcomplexes with non-empty intersection and $X = A \cup B$, consider the following square of inclusions:

$$\begin{array}{ccc} A \cap B & \longrightarrow & A \\ \downarrow & & \downarrow \\ B & \longrightarrow & X \end{array}$$

We have seen that such a square diagram is a homotopy pushout diagram. Now any inclusion map $W \hookrightarrow Z$ is k -connected if and only if (Z, W) is k -connected. So $(A, A \cap B)$ is k_1 -connected and (X, B) is k_2 -connected. Blaker's-Massey theorem implies that

$$\mathrm{hofiber}(A \cap B \rightarrow A) \rightarrow \mathrm{hofiber}(B, X)$$

is $(k_1 + k_2 - 1)$ -connected. But by definition we have an isomorphism $\pi_k(\mathrm{hofiber}(U \rightarrow V) \cong \pi_{k+1}(V, U)$. So we are really just saying that $\pi_k(A, A \cap B) \rightarrow \pi_k(X, B)$ given by the inclusion is $(k_1 + k_2)$ -connected.

7.4 The Dual Blakers-Massey Theorem for Squares

Theorem 7.4.1 (Dual Blakers-Massey Theorem for Squares) Let $X_0, X_1, X_2, X_{12} \in \mathbf{CGWH}$ be spaces such that the square

$$\begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_{12} \end{array}$$

is a homotopy pullback. Suppose the map $X_i \rightarrow X_{12}$ is k_i -connected for $i = 1, 2$. Then the diagram is $(k_1 + k_2 - 1)$ -cocartesian. Explicitly, this means that

$$\beta : \mathrm{hocolim}(X_1 \leftarrow X_0 \rightarrow X_2) \rightarrow X_{12}$$

is $(k_1 + k_2 - 1)$ -connected.

8 Homotopy n-Cubes

In algebraic topology, we have learnt about spaces, maps of spaces and maps of maps of spaces. We can say this in a more compact way. Namely, if we think of maps of maps of space as a square (2-cube), we can think of spaces as 0-cubes and maps of spaces as 1-cube. We have studied 2-cubes extensively under the guise of homotopy pullbacks and pushouts. We can now take this further and consider general n -cubes.

8.1 Maps between Homotopy Squares

8.2 n-Cubes of Spaces

Definition 8.2.1 (n-Cubes of Spaces) Let $n \in \mathbb{N}$. Let $P(n)$ denote the category of posets of the set $\{1, \dots, n\}$. An n -cube of spaces is a functor

$$X : P(n) \rightarrow \mathbf{CGWH}$$

An n -cube of based spaces is a functor $X : P(n) \rightarrow \mathbf{CGWH}_*$.

Explicitly, an n -cube of spaces $X : P(n) \rightarrow \mathbf{CGWH}$ consists of the following data.

- For each $S \subseteq \{1, \dots, n\}$ a space X_S
- For each $S \subseteq T$, a map $f_{S \subseteq T} : X_S \rightarrow X_T$ such that $f_{S \subseteq S} = 1_{X_S}$ and for all $R \subseteq S \subseteq T$, we have a commutative diagram

$$\begin{array}{ccc} X_R & \xrightarrow{f_{R \subseteq S}} & X_S \\ & \searrow f_{R \subseteq T} & \downarrow f_{S \subseteq T} \\ & & X_T \end{array}$$

Omit drawing composite arrows and omit drawing identities.

Also: punctured cubes def

Definition 8.2.2 (Cube of Cubes) An n -cube of m -cubes is a functor

$$X : P(n) \times P(m) \rightarrow \mathbf{CGWH}$$

Lemma 8.2.3 An n -cube of m -cubes X is precisely an $(n + m)$ -cube.

Definition 8.2.4 (Map of n-Cubes) Let $X, Y : P(n) \rightarrow \mathbf{CGWH}$ be n -cubes. A map of n -cubes is a natural transformation $F : X \rightarrow Y$ such that the assignment $Z : P(n + 1) \rightarrow \mathbf{CGWH}$ given by

$$Z(S) = \begin{cases} X(S) & \text{if } S \subseteq \{1, \dots, n\} \\ Y(S \setminus \{1, \dots, n + 1\}) & \text{if } \{1, \dots, n + 1\} \subseteq S \end{cases}$$

defines an $(n + 1)$ -cube.

objectwise (co)fibration, homotopy (weak) equivalence. homeomorphism

Definition 8.2.5 (Strongly Homotopy Cartesian) Let X be an n -cube of spaces. We say that X is strongly homotopy cartesian if each of its faces of dimension $n \geq 2$ is homotopy cartesian.

9 Homotopy Limits and Colimits

Let $X : \mathcal{J} \rightarrow \mathbf{Top}$ be a diagram of spaces. Denote the constant functor of the one point space by $\Delta* : \mathbf{X} \rightarrow \mathbf{Top}$. The data of the constant functor is given as follows.

- For each $I \in \mathcal{J}$, $\Delta*(I) = *$
- For each morphism $f : I \rightarrow J$ in $\mathcal{J}$, define $\Delta*(f) = \text{id}_*$.

Consider the set of all natural transformations $\Delta* \Rightarrow X$ denoted by $\text{Nat}(\Delta*, X)$. Now this set can inherit a subspace topology via the isomorphism (of sets)

$$\text{Nat}(\Delta*, X) \cong \prod_{I \in \mathcal{J}} \text{Hom}_{\mathbf{Top}}(*, X_I) \subset \prod_{I \in \mathcal{J}} X_I$$

There is in fact a canonical homeomorphism between the set of natural transformations and the limit of X .

Theorem 9.0.1 Let $X : \mathcal{J} \rightarrow \mathbf{Top}$ be a diagram of spaces. Then there is a canonical homeomorphism

$$\lim_{\mathcal{J}} X \cong \text{Nat}(\Delta*, X)$$

Ref: Cubical diagrams

Theorem 9.0.2 Let $X : \mathcal{J} \rightarrow \mathbf{Top}$ be a diagram of spaces. Then there is a canonical homeomorphism

$$\text{colim}_{\mathcal{J}} X \cong \frac{\coprod_{I \in \mathcal{J}} X_I}{\sim}$$

where $x \in X_I \sim y \in X_J$ if and only if there exists $f : I \rightarrow J$ such that $X(f)(x) = y$.

9.1 Homotopy Limits and Colimits

Let $\mathcal{J}$ be a small diagram. Let $I \in \mathcal{J}$ and denote $\mathcal{J}/I$ to be the overcategory with the distinguished object I . We can turn it into a topological space by constructing its classifying space

$$\mathcal{B}(\mathcal{J}/I) = |N(\mathcal{J}/I)|$$

which is the geometric realization of the nerve of $\mathcal{J}/I$. We aim to use the overcategory $\mathcal{J}/I$ to record homotopy information. Recall that the limit is canonically isomorphic to the equalizer of

$$f, g : \prod_{J \in \mathcal{J}} X_J \rightarrow \prod_{(\alpha : J \rightarrow I) \in \mathcal{J}} X_I$$

as follows.

- Define f to be the unique map such that $\pi_I \circ f = \pi_{\text{cod}(\alpha)}$ where both π are projections.
- Define g to be the unique map such that $\pi_{\text{cod}(\alpha)} \circ g = F(\alpha) \circ \pi_{\text{dom}(\alpha)}$

We aim to replace each space X_I by the space $\text{Map}(\mathcal{B}(\mathcal{J}/I), X_I)$ (which is why we work with **CGWH**). Indeed the hom space and X_I are homotopy equivalent since $\mathcal{B}(\mathcal{J}/I)$ is contractible. Notice that this is no longer a functor in I , but rather a bifunctor. This all works with a simplicial model category.

Definition 9.1.1 (Homotopy Limits) Let $\mathcal{C}$ be a simplicial model category. Let $X : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram. Define two maps

$$f, g : \prod_{J \in \mathcal{J}} \text{Map}(\mathcal{B}(\mathcal{J}/J), X_J) \rightarrow \prod_{(\alpha : J \rightarrow I) \in \mathcal{J}} \text{Map}(\mathcal{B}(\mathcal{J}/J), X_I)$$

as follows.

- Define f to be the unique map such that

$$\pi_I \circ f = \text{Map}(\mathcal{B}(\mathcal{J}/J), X(\alpha) : X_J \rightarrow X_I)$$

for any $\alpha : J \rightarrow I$ a morphism in $\mathcal{J}$.

- Define g to be the unique map such that

$$\pi_I \circ g = \text{Map}(\mathcal{B}(\mathcal{J}/X : \mathcal{J}/I \rightarrow \mathcal{J}/J), X_I)$$

for any $\alpha : J \rightarrow I$ a morphism in $\mathcal{J}$.

Define the homotopy limit $\text{holim} X$ of X to be the equalizer of the maps

$$\text{holim} X = \text{Eq}(f, g)$$

Let $\mathcal{J}$ be the small diagram consisting of two objects A, B and two non-trivial morphisms $f, g : A \rightarrow B$. Let us illustrate the homotopy limit of $\mathcal{J}$. The slice category $\mathcal{J}/A$ consists only of a single object corresponding to the identity map $\text{id}_A : A \rightarrow A$. The slice category $\mathcal{J}/B$ consists of 3 objects corresponding to id_B, f and g . Non-trivial morphisms are given as follows:

$$\begin{array}{ccc} & f & \\ & \searrow & \\ & \text{id}_B & \\ & \nearrow & \\ g & & \end{array}$$

Passing to the classifying space, $\mathcal{B}(\mathcal{J}/A)$ is just the one point space and $\mathcal{B}(\mathcal{J}/B)$ becomes the interval I .

Expanding things out show that

$$f, g : \text{Map}(\cdot, A) \times \text{Map}(\cdot, B)$$

Proposition 9.1.2 Let $X : \mathcal{J} \rightarrow \mathbf{CGWH}$ be a diagram. Then there is a natural transformation

$$\text{holim}_{\mathcal{J}} X \cong \text{Nat}(\mathcal{B}(\mathcal{J}/-), \mathcal{J} \rightarrow \mathbf{CGWH}, X)$$

Definition 9.1.3 (Homotopy Colimits) Let $\mathcal{C}$ be a simplicial model category. Let $X : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram. Define two maps

$$f, g : \coprod_{(\alpha: J \rightarrow I) \in \mathcal{J}} X_J \times \mathcal{B}((I/\mathcal{J})^{\text{op}}) \rightarrow \coprod_{J \in \mathcal{J}} X_J \times \mathcal{B}((J/\mathcal{J})^{\text{op}})$$

as follows.

- On each summand of the domain of f , define f to be the map

$$X(\alpha) \times \text{id}_{\mathcal{B}((I/\mathcal{J})^{\text{op}})} : X_J \times \mathcal{B}((I/\mathcal{J})^{\text{op}}) \longrightarrow X_I \times \mathcal{B}((I/\mathcal{J})^{\text{op}})$$

and then injecting the into the coproduct.

- On each summand of the domain of g , define g to be the map

$$\text{id}_{X_J} \times \mathcal{B}(\mathcal{J}/X)^{\text{op}} : X_J \times \mathcal{B}((I/\mathcal{J})^{\text{op}}) \longrightarrow X_J \times \mathcal{B}((J/\mathcal{J})^{\text{op}})$$

and then injecting into the coproduct.

Define the homotopy colimit $\text{hocolim} X$ of X to be the coequalizer of the maps

$$\text{hocolim} X = \text{Coeq}(f, g)$$

10 ???

10.1 The Relative Point of View

Up until this point, in algebraic topology we have asked questions relating to two spaces and tried to answer them. For instance, we can ask whether two spaces are homeomorphic, homotopy equivalent or weakly equivalent. We can also ask these questions in a relative setting, this involves considering maps of spaces as objects themselves, instead of just the spaces.

Definition 10.1.1 (Maps of Maps) Let X, Y, A, B be spaces. Let $f : X \rightarrow Y$ and $g : A \rightarrow B$ be maps. A map from f to g is a pair of maps $(\alpha : X \rightarrow A, \beta : Y \rightarrow B)$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \alpha \downarrow & & \downarrow \beta \\ A & \xrightarrow{g} & B \end{array}$$

Definition 10.1.2 (Homotopy from Maps to Maps) Let X, Y, A, B be spaces. Let $f : X \rightarrow Y$ and $g : A \rightarrow B$ be maps. Let (a, b) and (c, d) be two maps from f to g . We say that (a, b) and (c, d) are homotopic if there exists maps $H : X \times I \rightarrow Y$ and $K : A \times I \rightarrow B$ such that the following diagram

$$\begin{array}{ccc} X \times I & \xrightarrow{f \times \text{id}_I} & Y \times I \\ H \downarrow & & \downarrow K \\ A & \xrightarrow{g} & B \end{array}$$

commutes and the following are true.

- H is a homotopy from a to c
- K is a homotopy from b to d

We now restrict to the point of view where maps are over a fixed space Z (In other words we are considering objects of $\mathbf{Top}_Z$). A map of maps becomes the following data: Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be maps. A map from f to g is a map $h : X \rightarrow Y$ such that $g \circ h = f$. In other words the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{h} & Y \\ & \searrow f & \swarrow g \\ & Z & \end{array}$$

A homotopy then becomes the following data: Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be two maps. Let $h, k : X \rightarrow Y$ be two maps from f to g . We say that h and k are homotopic if there exists a homotopy $H : X \times I \rightarrow Y$ from h to k such that $H(-, t)$ for each t is a map over Z . This means that the following diagram commutes:

$$\begin{array}{ccc} X \times I & \xrightarrow{H} & Y \\ & \searrow f \circ \text{pr}_X & \swarrow g \\ & Z & \end{array}$$

We can now discuss homotopy equivalences on these maps.

Definition 10.1.3 (Fiber Homotopy Equivalence) Let X, Y, Z be spaces. Let $f : X \rightarrow Z$ and $g : Y \rightarrow Z$ be maps. We say that f and g are homotopy equivalent if there exists two maps $h : X \rightarrow Y$ and $k : Y \rightarrow X$ such that $k \circ h$ and $h \circ k$ are both homotopic to the identity over Z .

The reason that it is called a fiber homotopy equivalence is because it gives homotopy equivalences on fibers.